MILITARY SPECIFICATION

PARACHUTES, PERSONNEL, EMERGENCY ESCAPE

GENERAL DESIGN SPECIFICATION FOR

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the general performance, design, development, and inspection requirements for two types of emergency escape parachutes used for the recovery of personnel from aircraft under emergency escape conditions (see 1.2). This specification is not applicable to systems where personnel are recovered in a capsule or section of an aircraft.

1.2 Classification. Emergency escape personnel parachutes covered by this specification are of the following types (see 3.1):

Type I. Ejection seat parachute. The ejection seat parachute (or "recovery subsystem") is for use by an aircrewmember after exiting the aircraft via an ejection seat Aircrew Automated Escape System (AAES).

Type II. Bailout parachute. The bailout parachute is for use by an aircrewmember after manually exiting the aircraft, and may be worn during flight or donned immediately prior to egress or "bailout."

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering Standardization Department (SESD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC No. N3776

FSC 1670

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2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-C-320	Chromium Plating (Electrodeposited).
QQ-A-367	Aluminum Alloy Forgings.
DDD-L-20	Label: For Clothing, Equippage, and Tentage (General Use).

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DOD-D-1000	Drawings, Engineering and Associated Lists.
MIL-S-5002	Surfaces Treatments and Inorganic Coatings for Metal Surfaces of Weapons Systems.
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-P-5610	Parachute Assemblies and Major Sub-Assemblies, Packaging and Packing of.
MIL-E-6051	Electromagnetic Compatibility Requirements, Systems.
MIL-H-6088	Heat Treatment of Aluminum Alloys.
MIL-H-6875	Heat Treatment of Steel, Process for.
MIL-I-6903	Ink, Marking (For Parachutes and Other Textile Items).

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MIL-F-7179	Finishes and Coatings: Protection of Aerospace Weapons Systems, Structures and Parts; General Specification for.
MIL-F-7190	Forgings, Steel, for Aircraft and Special Ordnance Applications.
MIL-P-7567	Parachutes, Personnel, Detail Manufacturing Instructions for.
MIL-B-7883	Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aerospace Vehicles.
MIL-W-8611	Welding, Metal Arc and Gas, Steels, and Corrosion and Heat Resistant Alloys; Process for.
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-F-18264	Finishes: Organic, Weapons System, Application and Control of.
MIL-T-18303	Test Procedures; Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for.
MIL-S-18471	System, Aircrew Automated Escape, Ejection Seat Type: General Specification for.
MIL-A-21180	Aluminum-Alloy Castings, High Strength.
MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices.
MIL-A-22771	Aluminum Alloy Forgings, Heat Treated.
MIL-D-23615	Design and Evaluation of Cartridge Actuated Devices.

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MIL-I-23659	Initiators, Electric, General Design Specification for.
MIL-P-55110	Printed Wiring Boards.
MIL-H-81200	Heat Treatment of Titanium and Titanium Alloys.
MIL-C-83488	Coating, Aluminum, Ion Vapor Deposited.
MIL-S-85076	System, Automatic, Water Activated Parachute Release; Detail Specification for.
MIL-C-87115	Coating, Immersion Zinc Flake/ Chromate Dispersion.
MIL-P-87141	Parachutes.

STANDARDS

FEDERAL

FED-STD-H28	Screw-Thread Standards for Federal Services.
FED-STD-102	Preservation, Packaging, and Packing Levels.
FED-STD-191	Textile Test Methods.
FED-STD-595	Colors.
FED-STD-751	Stitches, Seams, and Stitchings.

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MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of.

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MIL-STD-470	Maintainability Program for Systems and Equipment.
MIL-STD-756	Reliability Modeling and Prediction.
MIL-STD-785	Reliability Program for Systems and Equipment, Development and Production.
MIL-STD-810	Environmental Test Methods and Engineering Guidelines.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-882	System Safety Program Requirements.
MIL-STD-889	Dissimilar Metals.
MIL-STD-1252	Inertia Friction Welding Process, Procedure and Performance Qualification.
MIL-STD-1385	Preclusion of Ordnance Hazards in Electromagnetic Fields, General Requirements for.
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment, and Facilities.
MIL-STD-1695	Environments, Working, Minimum Standards for.
DOD-STD-1866	Soldering Process, General (Non Electrical).
MIL-STD-2067	Aircrew Automated Escape Systems Reliability and Maintainability (R/M) Program Requirements for.
MIL-STD-2080	Maintenance Engineering, Planning, and Analysis for Aeronautical Systems, Equipment, and Support Equipment.
MIL-STD-2175	Castings, Classification and Inspection of.
MIL-STD-45662	Calibration Systems Requirements.

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HANDBOOKS

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MIL-HDBK-5	Metallic Materials and Elements for Aerospace Vehicle Structures.
MIL-HDBK-132	Protective Finishes.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

OTHER GOVERNMENT DOCUMENTS

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171-178	Transportation.

DRAWINGS

NAVAL WEAPONS CENTER (NWC)

SK 13-018-001-85	Parachute Volume Measurement Fixture.

PUBLICATIONS

CHIEF OF NAVAL OPERATIONS

OPNAVINST 4790.2	Naval Aviation Maintenance Program.

NAVAL AIR SYSTEMS COMMAND (NAVAIR)

NAVAIR 13-1-6.2	Aviation-Crew Systems Manual - Organization, Intermediate and Depot Level Maintenance With Illustrated Parts Breakdown for Emergency Personnel and Drogue Parachute Systems.
AS-4613	Application and Derating Requirements for Electronic Components, General Specification for.
WS-6536	Procedures and Requirements for Preparation and Soldering of Electrical Connections.

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NAVAL WEAPONS CENTER

NWC TP 6575	Parachute Recovery Systems Design Manual.

WRIGHT-PATTERSON AIR FORCE BASE

AFFDL-TR-78-151	Recovery Systems Design Guide.

INTER-RANGE INSTRUMENTATION GROUP (IRIG)

IRIG Document 106	Telemetry Standards.
IRIG Document 118	Test Methods for Telemetry Systems and Subsystems.

(Copies of specifications, standards, handbooks, other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Item description. Emergency escape personnel parachutes within the scope of this specification shall be fabric devices used by aircrewmembers for making a safe descent from an aircraft following ejection (Type I) or bailout (Type II). The Type I and Type II emergency escape personnel parachutes are collectively referred to hereinafter as the "parachute(s)." Unless otherwise specified herein, the following requirements apply to both the Type I and the Type II parachutes. (See 6.3 for applicable definitions.)

3.1.1 Specification sheets. The individual parachute requirements shall be as specified herein and in accordance with the applicable specification sheet (see 6.4). In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

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3.1.2 Design responsibility. Unless otherwise specified in the contract or purchase order, the contractor (see 6.3.1) shall be responsible for the design including, but not limited to, the selection of materials and manufacturing techniques, the integration of Government-furnished property (GFP) and contractor-furnished components into the parachute, and the integration of the parachute into the applicable prime system (see 6.2.1 and 6.3.2). NWC TP 6575, AFFDL-TR-78-151, and MIL-P-87141 may be used to aid in design definition, for development of the parachute, and for overall design guidance, with this specification and the associated specification sheet as the precedent documents. The final design and all elements of the design shall be approved by the contracting activity (see 6.3.3).

3.1.3 Interface. The parachute shall be designed to physically and functionally interface with the prime system (see 6.3.2) and other systems, as applicable. The interface control drawings and other engineering data required to define all physical and functional interfaces and to ensure parachute compatibility with the prime system and other applicable systems shall be as specified in the associated specification sheet (see 4.5.1 and 6.4).

3.2 Design characteristics. The parachute shall be designed to minimize complexity, maintenance, and handling, and to ensure maximum safety, performance, reliability, efficiency, and economical production. In addition to the requirements specified herein, the Type I parachute shall meet the requirements of MIL-S-18471.

3.2.1 Performance. The parachute shall meet the performance requirements of Table I. The requirements of Table I shall apply under standard conditions as defined in 6.3.4, with a 150- to 300-pound (lb) suspended load (see 6.3.5), and at a velocity at initiation of deployment up to 300 knots equivalent airspeed (KEAS). Unless otherwise specified in the associated specification sheet, the parachute shall be operable over the temperature range of -54 to +74 degrees Celsius (°C) (-65 to +165 degrees Fahrenheit (°F)) (see 6.4). The unique performance requirements for the parachute shall be specified in the associated specification sheet, but shall be within the performance envelope of Table I (see 4.5.2 and 6.4).

3.2.2 Flight control features. Unless otherwise specified in the associated specification sheet, the flight control system of the parachute shall include two modes: (1) the "hands-off" mode (see 3.2.2.1), and (2) the controlled gliding mode (see 3.2.2.2 and 6.4). The flight control system shall minimize the possibility of inadvertently stalling the canopy during maneuvering, and shall provide the capability of modulating the forward speed of the canopy using simultaneous left and right control inputs (see 4.5.3).

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TABLE I. Performance requirements.[1/]

Item	Parameter	Requirements[2/]
1	Vertical descent velocity	24 ft/s max
2	Horizontal velocity, hands-off mode[3/]	8 ft/s max
3	Oscillation from vertical	15 deg max
4	Time from initiation of deployment[4/] to point of last full open [5/] at a true airspeed at initiation of 100 ±10 ft/s and a density altitude[6/] of 8000 ±500 ft	2 sec max
5	Acceleration (vector sum) during deployment (opening shock) at a density altitude of 8000 ±500 ft and a velocity of 300 ±10 KEAS	15 g max
6	Rate-of-turn a. Hands-off mode[3/] b. Controlled gliding mode at full control stroke[7/]	5 deg/s max 45 to 90 deg/s

[1/] See 3.2.1, 6.3.4, and 6.3.5.

[2/] The specified requirements shall be the limiting single values for any individual test. However, when a single value is statistically derived from continuous or multiple test data (such as for Item 1), the derived single value to be compared to the requirement shall be the upper two-σ value when compared to a maximum and the lower two-σ value when compared to a minimum.

[3/] See 3.2.2.1.

[4/] See 6.3.6.

[5/] See 6.3.7.

[6/] See 6.3.8.

[7/] See 3.2.2.2 and 6.3.9.

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3.2.2.1 Hands-off mode. The parachute shall include a low-gliding (see 6.3.10) hands-off mode to accommodate an injured or unconscious aircrewmember. The parachute shall deploy in the hands-off mode with selection of the controlled gliding capability at the option of the aircrewmember. In the hands-off mode, the rate-of-descent shall be in accordance with Table I, Items 1 and 2, and the rate-of-turn shall be in accordance with Table I, Item 6a (see 4.5.3).

3.2.2.2 Controlled gliding mode. The controlled gliding mode shall provide a horizontal velocity in excess of 8 feet per second (ft/s), or as specified in the associated specification sheet (see 6.4). The rate-of-turn in the controlled gliding mode shall be in accordance with Table I, Item 6b. The controlled gliding mode shall be capable of manual activation/selection and manual deactivation/deselection by the aircrewmember. Within a 200-foot altitude loss following deactivation/deselection by the aircrewmember, the parachute shall return to the hands-off mode of 3.2.2.1 (see 4.5.3).

3.2.3 Manually-operated features.

3.2.3.1 Container/pack manual opening system. The Type II parachute shall include a manually-activated container/pack opening system designed for operation by either hand. When the Type I parachute includes a manually-activated container/pack opening system it shall also be designed for operation by either hand (see 4.5.4.1).

3.2.3.2 Manual operation force. For both the Type I and Type II parachutes, the container/pack manual opening system of 3.2.3.1 shall be designed for operation with a pull force not less than 10 nor greater than 27 pound-force (lbf). The minimum and maximum forces required for an aircrewmember to manually activate other design features of the parachute which require a pull-force to invoke operation shall be as specified in the associated specification sheet (see 6.4 and 4.5.4.2).

3.2.4 Canopy/risers release system. The parachute shall incorporate a canopy/risers release system to detach the canopy or risers from the aircrewmember. The canopy/risers release system shall be designed to preclude inadvertent operation and shall have not greater than two points of release, each operable with either hand and by two separate and different motions. For the Type I parachute, the canopy/risers release system shall also function automatically upon contact with seawater in accordance with MIL-S-85076 (see 4.5.5).

3.2.4.1 Inadvertent canopy/risers release. The parachute shall be designed to maintain the integrity of the main canopy in the event of the inadvertent functioning of one side of a dual canopy/risers release system (see 3.2.4). The parachute shall sustain no damage nor exhibit performance degradation worse than for a marginal failure as defined in 3.2.14.3 (see 4.5.5.1).

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3.2.5 Canopy overload. With a 300-lb suspended load and upon inflation of the main canopy, the parachute shall be capable of withstanding a total riser force of 6000 lbf minimum and a single riser force of 4500 lbf minimum. The parachute shall sustain no damage nor exhibit performance degradation worse than for a critical failure as defined in 3.2.14.3 (see 4.5.6).

3.2.6 Canopy deflation features. To hinder the parachute's proclivity to drag the aircrewmember across the surface of the water, the parachute shall incorporate features (such as water deflation pockets) which will aid in canopy deflation after a water landing (see 4.5.7).

3.2.7 Harness. When the design of the parachute includes the harness, the following requirements shall apply (see 4.5.8):

a. The harness shall preclude injury to the aircrewmember during transfer of the opening forces from the main canopy to the aircrewmember, and shall safely and securely support the aircrewmember during all phases of operation without damage to the harness.

b. Webbing straps shall not slip greater than 0.5 inch through the hardware when subjected to a pull force from 0 to 6500 lbf.

c. For Type I parachutes, the harness shall interface with the ejection seat AAES specified in the associated specification sheet (see 6.4).

3.2.8 Twisted-line opening capability. With the suspension lines twisted together 360 degrees three times within 30 inches beginning immediately below the deployment bag locking or emergence point, or beginning immediately below the canopy skirt for the parachute not incorporating a deployment bag, and with the parachute then packed in the approved manner, the parachute shall deploy as designed, the canopy shall fully inflate, and the suspension lines shall completely untwist within 500 ±50 feet following initiation of deployment at a velocity not greater than 100 KEAS and with a 300-lb suspended load (see 4.5.9).

3.2.9 Ballistics. Prior to implementation, all ballistic components/devices proposed for use in the design shall be approved for use by the contracting activity (see 4.5.10).

3.2.9.1 Cartridge-actuated devices. Cartridge-actuated devices used in the design shall meet the requirements of MIL-D-23615. Internal cartridges for the cartridge-actuated devices shall meet the requirements of MIL-D-21625 (see 4.5.10.1).

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3.2.9.2 Electric initiators. Electric initiators used in the cartridge-actuated devices and elsewhere in the design shall meet the requirements of MIL-I-23659 (see 4.5.10.2).

3.2.9.3 Gas-actuated devices. Gas-actuated devices used in the design shall meet the requirements of MIL-D-23615 (see 4.5.10.3).

3.2.10 Physical characteristics.

3.2.10.1 Weight. The maximum allowable weight shall be as specified in the associated specification sheet (see 4.5.11.1 and 6.4).

3.2.10.2 Dimensions/volume. Dimension and volume requirements shall be as specified in the associated specification sheet (see 4.5.11.2 and 6.4).

3.2.10.3 Color. The color scheme of the parachute, other than the canopy, shall be determined by the contracting activity and shall be as specified in the associated specification sheet (see 6.4). Unless otherwise specified in the specification sheet, the canopy shall be color-coded and the colors shall be sequenced as follows (see 4.5.11.3 and 6.4):

a. 36 percent international orange in accordance with FED-STD-595, Color Number 12246; then

b. 36 percent natural white; then

c. 14 percent sand shade in accordance with FED-STD-595, Color Number 30450; then

d. 14 percent olive green in accordance with FED-STD-595, Color Number 34092.

3.2.11 Rigging and packing. The parachute shall be packed in a manner which prevents deploying parachute components from striking the aircrewmember and from damaging garments or equipment. Special rigging and packing tooling peculiar to a specific design shall be kept to a minimum. The rigging and packing procedures shall be documented by the contractor and, when specified in the contract or purchase order, shall be included in the data package as part of the design disclosure (see 6.2.2). Prior to live jump-testing, the rigging and packing procedures shall be validated/certified by the certification group specified in the contract or purchase order (see 6.2.1). When specified in the contract or purchase order, the validation/certification shall be documented in a certification/data report (see 6.2.2). Rigging and packing procedures for current in-service parachutes are specified in NAVAIR 13-1-6.2 which may be used for guidance (see 4.5.12).

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3.2.12 Environmental conditions.

3.2.12.1 Rain. The parachute, in the packed condition (see 6.3.11), shall meet the requirements of Table I after exposure to a rain environment consisting of a rainfall rate up to 0.8 millimeters per minute (mm/min), droplet sizes from 2.0 to 4.5 millimeters (mm), wind velocities up to 64 kilometers per hour (km/hr), and water temperatures from +5 to +45°C (+40 to +115°F) for 30 minutes per parachute container/pack side (see 4.5.13.1).

3.2.12.2 Salt fog. The parachute, in the packed condition, shall meet the requirements of Table I after exposure to a salt fog environment consisting of salt concentrations up to 5 percent, atomized at 2.8 liters of salt solution per 0.28 cubic meters (m^3) and temperatures up to +35°C (+95°F) for 48 hours (see 4.5.13.2).

3.2.12.3 Sand and dust. The parachute, in the packed condition, shall meet the requirements of Table I after exposure to a sand and dust environment consisting of temperatures up to +49°C (+120°F), air velocities up to 5700 feet per minute (ft/min), dust concentrations up to 0.5 grams per cubic foot (g/ft^3) for 12 hours minimum, and sand concentrations up to 0.0773 g/ft^3 for 1.5 hours (see 4.5.13.3).

3.2.12.4 Pressure/altitude cycling. The parachute, in the packed condition, shall meet the requirements of Table I after exposure to a pressure/altitude environment cycling from 5 pound-force per square inch absolute (psia) for 1 minute minimum to 15 psia for 1 minute minimum (one cycle) for 5000 cycles (see 4.5.13.4).

3.2.12.5 High temperature/humidity. The parachute, in the packed condition, shall meet the requirements of Table I after exposure to a high temperature environment of +74°C (+165°F) with the humidity levels at 5 percent relative humidity (RH) maximum for 500 hours and 90 percent RH minimum for 500 hours (see 4.5.13.5).

3.2.12.6 High temperature cycling. The parachute, in the packed condition, shall meet the requirements of Table I after exposure to a high temperature environment cycling from +20 to +74°C (+68 to +165°F) in 1 hour, remaining at +74°C (+165°F) for 5 hours, then back down to +20°C (+68°F) in 1 hour and remaining at +20°C (+68°F) for 3 hours (one cycle) for 100 cycles (see 4.5.13.6).

3.2.12.7 Temperature/humidity shock. The parachute, in the packed condition, shall meet the requirements of Table I after exposure to temperature/humidity shock beginning at +74°C (+165°F) and 90 percent RH for 2 hours, then, within one minute, down to -54°C (-65°F) at 5 percent RH for 2 hours (one cycle) for 50 cycles (see 4.5.13.7).

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3.2.12.8 Solar radiation. The parachute, in the packed condition, shall meet the requirements of Table I after continuous exposure to noonday sun solar radiation equivalent to 1120 watts per square meter (W/m^2) for 1440 hours (see 4.5.13.8).

3.2.12.9 Fungus. When the design includes fungi-nutrient materials (see 3.3.1.2), the parachute, in the packed condition and the exposed condition (see 6.3.12), shall not support fungi growth when exposed to United States Department of Agriculture (USDA) fungi QM 386, QM 380, QM 432, QM 474, and QM 459 for 672 continuous hours, and shall meet the requirements of Table I after exposure (see 4.5.13.9).

3.2.12.10 Acceleration, shock, and vibration. The parachute, in the packed condition, shall meet the requirements of Table I after exposure to the acceleration, shock, and vibration levels specified in the associated specification sheet which have been derived from the prime system detail specification (see 4.5.13.10 and 6.4).

3.2.12.11 Depressurization. The parachute, in the packed condition, shall meet the requirements of Table I after exposure to rapid depressurization from 15 to 1 psia within 0.1 second (see 4.5.13.11).

3.2.13 Service life and shelf/total life. The service life (see 6.3.13) and shelf/total life (see 6.3.14) of the parachute shall be as specified in the associated specification sheet (see 4.5.14 and 6.4).

3.2.14 Reliability. The parachute shall demonstrate a reliability of 0.98 minimum at the 90 percent confidence level when tested as specified herein. These requirements assume that system sequencing of the parachute has been designed such that any component or function that fails during operation shall "fail safe," in that redundant, standby devices will automatically continue safe performance through mission completion (see 4.5.15).

3.2.14.1 Reliability program. When specified in the contract or purchase order, a reliability program shall be planned, implemented, and reported by the contractor in accordance with MIL-STD-2067 (for Type I only) and MIL-STD-785, Tasks 101, 102, 104, 204, 208, 209, and 302 (see 6.2.2). The reliability program plan shall be prepared for use by the contracting activity to assist in managing an effective reliability program and to evaluate the contractor's approach to, understanding of, and execution of his reliability tasks, his depth of planning to ensure that his procedures for implementing and controlling reliability tasks are adequate, and his organizational structure to ensure that appropriate attention will be focused on reliability activities and problems.

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3.2.14.2 Predicted reliability. Using the methods of MIL-STD-756 and the guidelines of MIL-STD-785, Task 203, or MIL-STD-2067 (for Type I only), a reliability prediction shall be performed by the contractor to estimate the basic, mission, and storage reliability of the parachute. The prediction shall be prepared for use by the contracting activity to determine whether these reliability requirements can be achieved with the proposed design and in making program decisions regarding the feasibility and adequacy of a new design approach. When specified in the contract or purchase order, all effort performed shall be documented in a report prepared in accordance with MIL-STD-756 (see 6.2.2).

3.2.14.3 Failure modes, effects, and criticality analysis (FMECA). When specified in the contract or purchase order, an FMECA shall be performed and documented by the contractor in accordance with MIL-STD-2067 (for Type I only) and MIL-STD-785, Task 204 (see 6.2.2). The FMECA shall be performed on each level of assembly for each required environmental condition and for each operational mode, and shall identify and evaluate all failure modes. In addition, for each failure mode, the likelihood of occurrence shall be determined for the failure identified, a determination shall be made as to its effect and criticality upon the component or subsystem in question; and of the ultimate significance of this effect upon the parachute performance, reliability, maintainability, and safety. These analyses shall include a description of the factors inherent in the design, or in the quality assurance program, that will minimize the probability of occurrence of those failures having the most significant potential adverse effect on system performance, reliability, maintainability, and safety. The criticality of the effects of each identified failure mode upon specific modes of parachute operation shall be classified or ranked into the following categories for a parachute operating at the extreme required performance envelopes:

a. Catastrophic. A catastrophic failure is defined as a design weakness, malfunction, or system damage that causes the parachute to fail to meet any of the performance requirements of Table I by greater than 50 percent.

b. Critical. A critical failure is defined as a design weakness, malfunction, or system damage that causes the parachute to fail to meet any of the performance requirements of Table I by less than 50 percent with a resulting failure that cannot be counteracted or controlled by an uninjured aircrewmember.

c. Marginal. A marginal failure is defined as a design weakness, malfunction, or system damage that causes the parachute to fail to meet any of the performance requirements of Table I by less than 50 percent with a resulting failure that can be counteracted or controlled by an uninjured aircrewmember.

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d. Negligible. A negligible failure is defined as a design weakness, malfunction, or system damage that does not prevent the parachute from meeting any of the performance requirements of Table I.

3.2.15 Maintainability. (See 4.5.16.)

3.2.15.1 Maintainability, nonsealed parachute. The nonsealed parachute (see 6.3.15) shall be designed to be maintained at the organizational (O) and intermediate (I) levels of maintenance and shall meet the following minimum maintainability requirements:

a. Based upon a flying hour rate of 35 hours per month, the direct maintenance man hour per flight hour (DMMH/FH) at the O and I levels for corrective and preventative maintenance shall be not greater than 0.05, including time for preparation of components to be installed in the parachute.

b. The mean-time-to-repair (MTTR) shall be not greater than 0.40 hour.

c. The maximum corrective maintenance time ($M_{max_{ct}}$) for the 95th percentile individual maintenance action, ranked on the basis of corrective times, shall be not greater than 1.2 hours, excluding time for rigging and packing.

d. At least 95 percent of the total maintenance actions can be performed by maintenance personnel within the Navy Aviation Structural Mechanic (AME) or Aircrew Survival Equipmentman (PR) rating at pay grade E-4 or below, or the civilian equivalent.

3.2.15.2 Maintainability, sealed parachute. Except for external components, the sealed parachute (see 6.3.16) shall be designed for zero maintenance at the O and I levels. The replaceable textile and hardware components external to the sealed parachute shall be designed to be maintained at the O and I levels in accordance with 3.2.15.1.

3.2.15.3 Maintainability program. When specified in the contract or purchase order, the contractor shall establish and maintain an effective maintainability program that is planned, integrated, prepared, and reported in accordance with MIL-STD-2067 (for Type I only) and MIL-STD-470, Task 101 (see 6.2.2). The maintainability program plan shall be developed in conjunction with other design and development functions to permit the most cost-effective achievement of overall program objectives. The maintainability program plan will be used by the contracting activity to improve operational readiness, reduce maintenance manpower needs, reduce life cycle costs, and for effective management through the data provided.

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3.2.16 Transportability. The parachute shall be designed to be transported by any appropriate transportation method suitable for explosives, if applicable, and which conforms to 49 CFR 171-178 (see 4.5.25 and Section 5).

3.3 Construction.

3.3.1 Materials, processes, and parts. (See 4.5.17.)

3.3.1.1 Materials and standard parts selection. Materials, including textiles, and standard parts shall conform to Government specifications and standards as specified herein or in the associated specification sheet (see 6.4). Materials or parts not covered by Government specifications or standards shall be suitable for the purpose intended, shall be of the best quality and the lightest practical weight, and shall be approved by the contracting activity prior to use. The following order of precedence shall be used for the selection of materials and standard parts:

a. Materials and parts specified in the contract or purchase order (see 6.2.1).

b. Materials and parts specified in this specification or in the associated specification sheet (see 6.4).

c. Other materials and parts covered by military specifications and standards.

d. Materials and parts specified in NWC TP 6575.

e. Materials and parts specified in AFFDL-TR-78-151.

f. Other materials and parts covered by industry specifications and standards and approved by the contracting activity.

g. Other materials and parts not covered by Government or industry specifications and standards, but approved by the contracting activity.

3.3.1.2 Fungus-proof materials. Materials which are nutrients for fungi shall not be used without prior approval from the contracting activity. When such materials must be used and approval for use has been obtained, they shall be treated with a fungicidal agent which has been approved by the contracting activity, and shall meet the requirements of 3.2.12.9.

3.3.1.3 Reclaimed, recycled, or reused parts and materials. No reclaimed, recycled, or reused parts or materials shall be used, except for raw steel, aluminum, or other metals which are the result of normal recycling processes.

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3.3.1.4 Toxic or corrosive emissions. Materials shall not generate toxic or corrosive emissions when exposed to temperatures from -54 to +74°C (-65 to +165°F). In the event of any uncertainty with regard to this requirement, the contractor shall immediately notify the contracting activity.

3.3.1.5 Textiles. Textiles shall be resistant to damage caused from exposure to solar radiation, atmospheric contaminants, including smog, and shipboard environments. Textiles that become unserviceable under conditions of service- or storage-use shall not be used.

3.3.1.5.1 Age of textiles. The contractor shall have available for Government review, records from the textile manufacturer which validate the age of all textiles used in the design. The date of manufacture and the name of the textile manufacturer for each material lot shall be recorded on the inspection record (see 4.4.1.3.2) for each parachute produced. The maximum allowable age shall be as specified in the associated specification sheet (see 6.4).

3.3.1.6 Electronic components. The use of electronic components in the design shall be approved by the contracting activity prior to implementation. Electronic components shall be selected from existing DOD standards and specifications to the fullest extent possible. For electronic components not covered by DOD standards and specifications, requirements comparable to the closest DOD standard component shall be specified in the associated specification sheet or on the applicable specification control drawing. Component application and derating shall be in accordance with AS-4613. Component quality levels shall be capable of achieving the performance requirements of Table I at the reliability level of 3.2.14.

3.3.1.6.1 Printed wiring boards. Printed wiring boards shall be in accordance with MIL-P-55110.

3.3.1.6.2 Wiring. Wiring and terminal design shall include protection against environmental and mechanical hazards to prevent corrosion and shorting of circuits. Electrical connectors shall be capable of being connected and disconnected and the cables flexed without damage within the temperature range of -54 to +74°C (-65 to +165°F).

3.3.1.7 Nonmetallic parts. Nonmetallic parts shall be of a type that minimize deterioration caused by abrasion or exposure to solar radiation, micro-organisms, moisture, heat, fuel, hydraulic and lubricating oil, grease, and salt spray. Nonmetallic materials shall have chemical properties appropriate for the environmental conditions specified herein. In addition, any nonmetallic parts used in the design shall not outgas corrosively, creep under an applied load, nor crack due to aging. Aging of nonmetallic parts for up to ten years shall not cause performance degradation in any service-use environment.

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3.3.1.8 Metallic parts. The selection of metals and their properties shall be in accordance with MIL-HDBK-5. Magnesium and magnesium alloys shall not be used.

3.3.1.8.1 Cables. Metal cables may be used in the design contingent upon approval by the contracting activity.

3.3.1.8.2 Dissimilar metals and organic matrix composites. Dissimilar metals and organic matrix composites, as defined and classified in MIL-STD-889, shall not be used unless suitably protected against electrolytic and galvanic corrosion in accordance with MIL-F-7179.

3.3.1.8.3 Corrosion protection. All metal parts shall be the corrosion resistant type, or shall be treated in accordance with MIL-F-7179 or as specified in 3.3.1.8.4, 3.3.1.8.5, or 3.3.1.8.6, as applicable, to render them corrosion resistant.

3.3.1.8.4 Finishes. The selection of protective finishes shall be in accordance with MIL-HDBK-132. Protective coatings and finishes shall be acceptable as moisture, fungus, and corrosion preventatives, and shall not crack, chip, nor scale during the normal course of inspection and operation or during exposure to the environmental conditions specified herein. Except as otherwise specified herein, inorganic coatings, surface treatments, and finishes shall be in accordance with MIL-S-5002. Application and control of organic finishes shall be in accordance with MIL-F-18264.

3.3.1.8.5 Anodizing, coating, and plating of aluminum parts. Aluminum and aluminum alloy parts not subject to wear, abrasion, erosion, or corrosion conditions shall be either anodized in accordance with MIL-A-8625, Type II, anodic coating, or treated with a chemical conversion coating in accordance with MIL-C-5541. Parts subject to normal wear, abrasion, erosion, or corrosion conditions shall be anodized in accordance with MIL-A-8625, Type II, anodic coating. Parts subject to severe wear, except for those normally reworked during overhaul, shall be anodized in accordance with MIL-A-8625, Type III. Parts subject to severe wear and which will normally be reworked during overhaul shall be chromium-plated in accordance with QQ-C-320, Class 2. Intermediate nickel plating may be used, as required, to render parts corrosion resistant.

3.3.1.8.6 Plating of steel parts. Steel parts in contact with aluminum or aluminum alloy parts shall be protected in accordance with MIL-C-87115 or coated with aluminum by ion vapor deposition in accordance with MIL-C-83488, Type II, Class 1.

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3.3.1.8.7 Heat treatment. Heat treatment of aluminum and aluminum alloy parts shall be in accordance with MIL-H-6088. Heat treatment of steel and steel alloy parts shall be in accordance with MIL-H-6875. Heat treatment of titanium and titanium alloy parts shall be in accordance with MIL-H-81200.

3.3.1.8.8 Castings. Castings shall not be used for parts which are stressed more than 50 percent of the material yield strength. Ferrous and nonferrous raw metal castings used in the design shall be in accordance with MIL-STD-2175. Aluminum alloy castings shall be in accordance with MIL-A-21180.

3.3.1.8.9 Forgings. Steel forgings shall be in accordance with MIL-F-7190, Grade A. Aluminum forgings shall be in accordance with QQ-A-367 and MIL-A-22771.

3.3.1.8.10 Welding. Welding shall be in accordance with MIL-W-8611 or MIL-STD-1252, as applicable.

3.3.1.8.11 Brazing. Brazing shall be in accordance with MIL-B-7883.

3.3.1.9 Soldering.

3.3.1.9.1 Mechanical soldering. Mechanical soldering shall be in accordance with DOD-STD-1866.

3.3.1.9.2 Electrical soldering. Procedures and requirements for preparation and soldering of electrical connections shall be in accordance with WS-6536.

3.3.1.10 Threads. Threads and the go/no-go gauging practice shall be in accordance with FED-STD-H28.

3.3.2 Fabrication procedures, textile components. Except as otherwise specified in the associated specification sheet or herein, detail manufacturing instructions shall be in accordance with MIL-P-7567 (see 6.4). Fabrication procedures shall be documented by the contractor and, when specified in the contract or purchase order, shall be included in the data package as part of the design disclosure (see 4.5.18 and 6.2.2).

3.3.2.1 Stitches, seams, and stitchings. Except as otherwise approved by the contracting activity, stitches, seams, and stitchings shall be in accordance with FED-STD-751.

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3.3.2.2 Suspension lines. Unless otherwise prescribed by unique design requirements, all suspension lines shall be continuous and without splice or knots. All lines for one canopy shall be made from the same continuous length of cord. Lines damaged during parachute construction shall be replaced with lines from the same line production lot whenever possible and, in any case, by lines from the same manufacturer.

3.3.2.2.1 Tensioning of suspension lines. After 24 hours of ambient conditioning of suspension line material under zero tension and at a temperature of 20 ±3°C (68 ±5°F) and RH from 40 to 80 percent, the suspension lines shall be marked for cutting under applied tension as specified in Table II. Within 30 ±5 seconds after applying tension, the lines shall be either marked and released from tension or locked in place under the applicable tension until marking can be performed. Lines tensioned on multiple pulley systems shall be measured and marked as one continuous length and shall not be anchored at any intermediate points between the ends.

TABLE II. Tension requirements for marking and cutting suspension lines.

Rated tensile strength of suspension line material, lb[1/]	Tension to be applied, lbf
100 to 499	10.0 ±0.5
500 to 1000	20.0 ±1.0
greater than 1000	40.0 ±2.0

1/ Ranges are inclusive.

3.3.3 Nameplates and product marking. Non-GFP equipment, assemblies, subassemblies, and components shall be marked for identification in accordance with MIL-STD-130 and as specified herein and in the associated specification sheet (see 6.4). (GFP components will be marked by the original manufacturer in accordance with their respective drawings prior to delivery.) The date of manufacture shall be shown for each fabric component, and shall be indicated by the month and year of delivery to the contracting activity. Space shall also be provided for entering the date the equipment, assembly, or subassembly is placed in service, to be filled in by the using activity. Parachutes manufactured for use by agencies other than the U.S. Government shall be identified and marked as specified in the contract or purchase order (see 6.2.1). Unless otherwise specified in the associated specification sheet, marking ink shall be in accordance with MIL-I-6903, Type I, II, III, or IV, as applicable, color generic black or blue (see 6.4). All marking shall be applied in a manner which will ensure clear legibility throughout the service life of the parachute (see 4.5.19).

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3.3.3.1 Marking of canopy. Unless otherwise specified in the associated specification sheet, all canopies, including pilot and drogue, shall be marked as specified on Figure 1 or 2, as applicable (see 6.4). Marking characters shall be not less than 0.5 inch high. Where applicable, the type designation shall precede the drawing number. For round canopies less than 36 inches in diameter, the date of manufacture and serial number may be specified only once (instead of twice), but in the top center gore as shown on Figure 1; and marking characters may be reduced to not less than 0.375 inch high, provided legibility is not adversely affected.

3.3.3.2 Marking of hardware. The part number shall be marked on each metal part large enough to allow legible marking. If possible, the manufacturer's Federal Supply Code for Manufacturers (FSCM) or symbol and year of manufacture shall also be marked on each metal part. Marking shall be applied by etching or die-stamping into the metal before the part is anodized or plated, or, for larger parts, by a preprinted metal-foil label.

3.3.3.3 Marking of harness. When the harness is included in the design and unless otherwise specified in the associated specification sheet, a label in accordance with DDD-L-20, Type IV, Class 11, shall be affixed to each harness (see 6.4). The label shall contain the following information: Design activity code identification number, part/drawing number, contract number, manufacturer's FSCM or symbol, date of manufacture, and space for date placed in service to be filled in by the using activity.

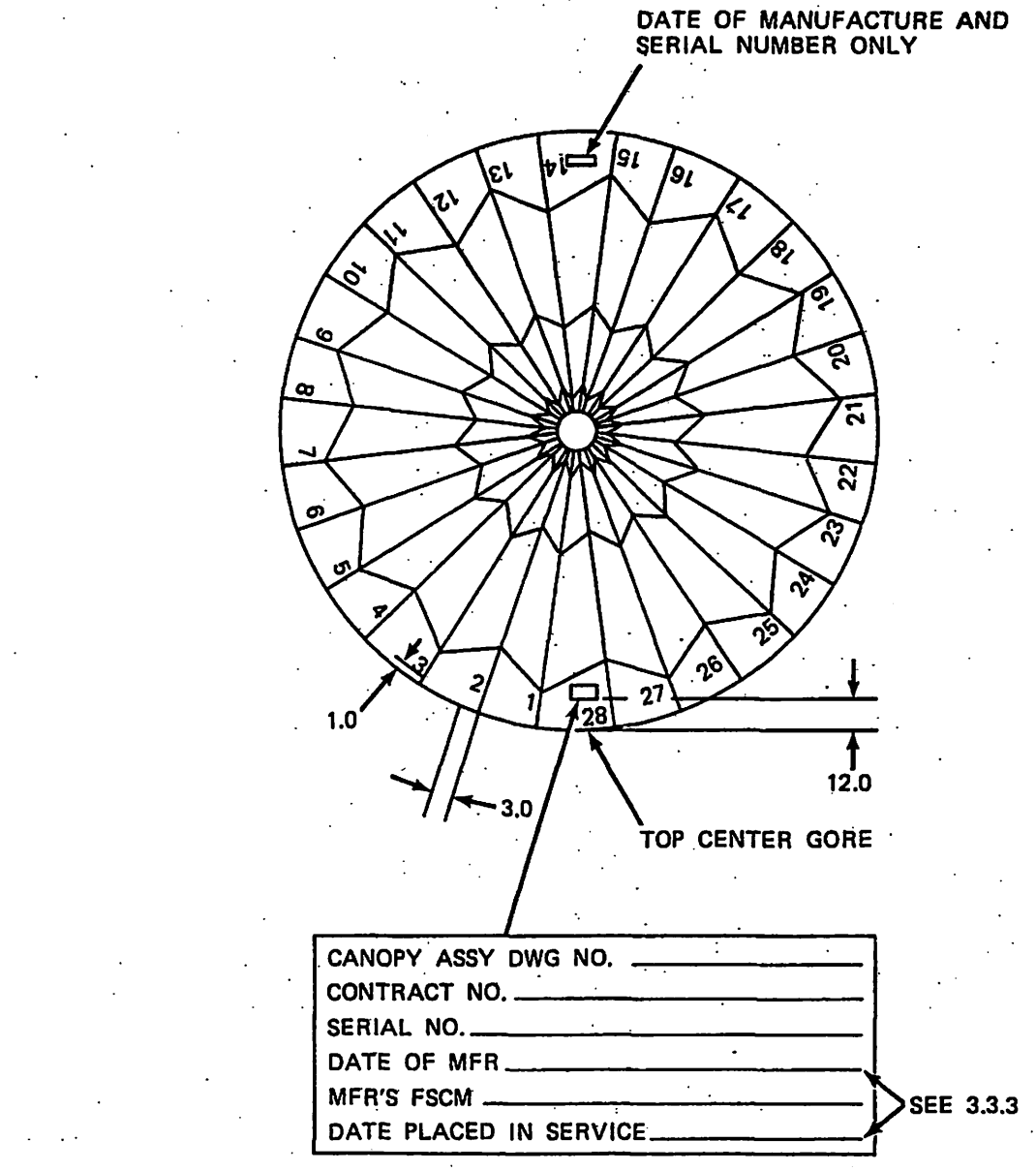
3.3.3.4 Marking of risers. When risers are included in the design which are not an integral part of the harness, each riser shall be marked as specified in 3.3.3.3 and shall be clearly identified as the right or left riser, if applicable.

3.3.3.5 Marking of cross-connector straps. When cross-connector straps are included in the design which are not an integral part of the riser or harness, each cross-connector strap shall be marked as specified in 3.3.3.3 and shall be clearly identified as the front or back strap, if applicable.

3.3.3.6 Marking of deployment bag. When a deployment bag is included in the design, the bag shall be marked as specified in 3.3.3.3.

3.3.3.7 Marking of container/pack. The outside of the container/pack shall be marked as specified in 3.3.3.3. If the parachute contains ballistic or pyrotechnic components, appropriate warning labels shall be affixed to both the outside and the inside of the container/pack and shall be defined in the associated specification sheet (see 6.4). In addition, the information shown on Figure 3 shall be marked on the outside of the container/pack, separate and apart from the label showing the container/pack part number and manufacturing information.

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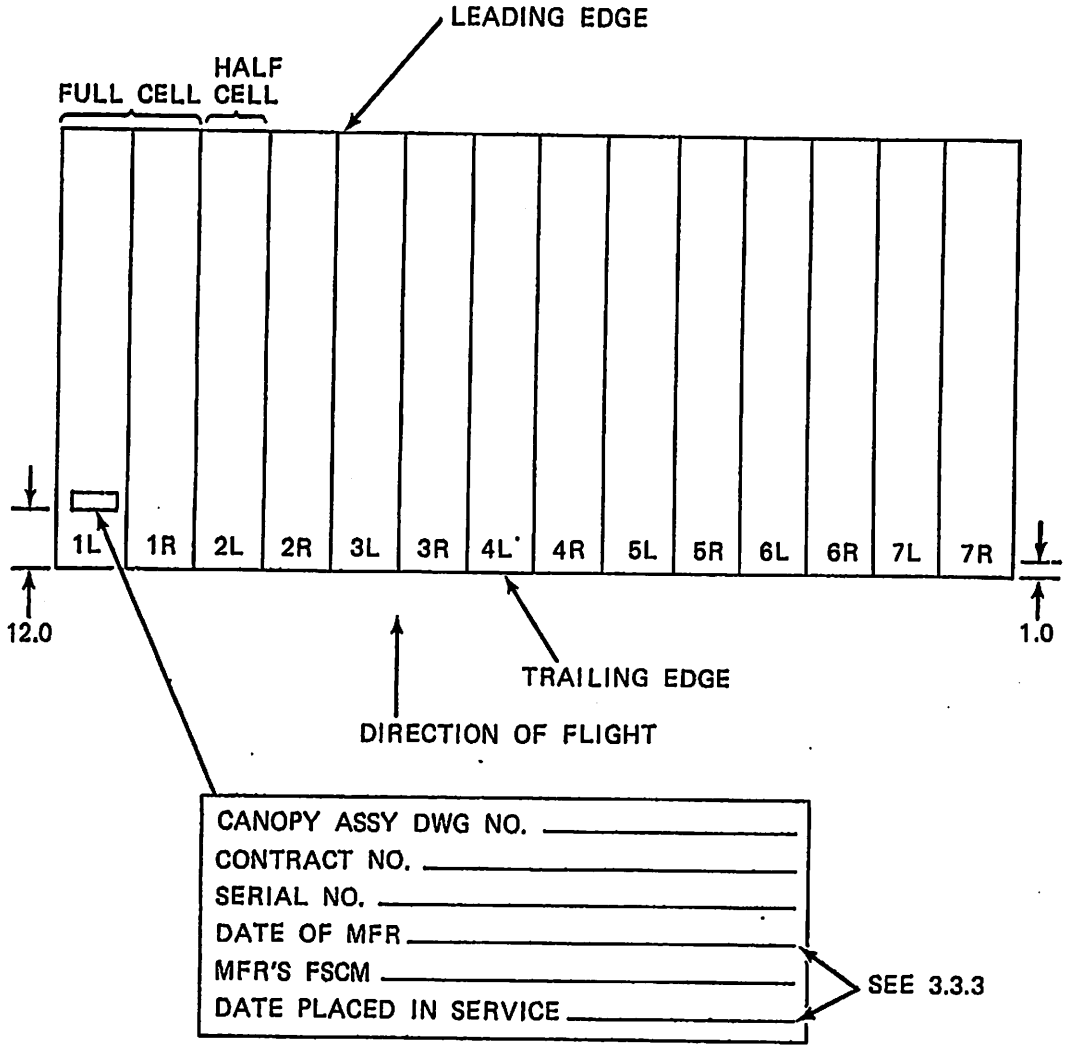


NOTES:

1. DIMENSIONS ARE IN INCHES, ±0.25 INCH
2. AS SHOWN ABOVE, GORES SHALL BE NUMBERED/MARKED IN A CLOCKWISE DIRECTION, ON THE RIGHT SIDE OF EACH GORE, AND ALONG THE HEM OF THE CANOPY.
3. FOR CANOPIES LESS THAN 36 INCHES IN DIAMETER, DATE OF MANUFACTURE AND SERIAL NUMBER NEED NOT BE REPEATED AS SHOWN IN GORE 14 ABOVE (SEE 3.3.3.1).
4. CANOPY DIAGRAM IS FOR REFERENCE ONLY. CANOPY SHAPE AND NUMBER OF GORES MAY VARY ACCORDING TO DESIGN REQUIREMENTS.

FIGURE 1. Round canopy marking requirements, view looking down on outside of canopy.

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NOTES:

1. DIMENSIONS ARE IN INCHES, ±0.25 INCH.
2. AS SHOWN ABOVE, CELLS SHALL BE NUMBERED/MARKED FROM LEFT TO RIGHT ALONG THE TRAILING EDGE OF THE CANOPY REFERENCED TO DIRECTION OF FLIGHT. CELL NUMBERS SHALL BE MARKED ON THE UPPER SURFACE IN THE CENTER OF EACH HALF CELL USING "L" (LEFT) AND "R" (RIGHT) TO DESIGNATE THE APPROPRIATE HALF CELL.
3. CANOPY DIAGRAM IS FOR REFERENCE ONLY. CANOPY SHAPE AND NUMBER OF CELLS MAY VARY ACCORDING TO DESIGN REQUIREMENTS.

FIGURE 2. Ram-air-inflated canopy marking requirements, view looking down on upper surface of canopy.

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NOMENCLATURE:
DESIGN ACTIVITY CODE IDENT:
PARACHUTE ASSY PART/DWG NO.:
PARACHUTE ASSY CONTRACT NO.:
MANUFACTURER'S FSCM:
ASSEMBLY ACTIVITY:
DATE OF ASSEMBLY:
CANOPY SERIAL NO.:
DATE CANOPY PLACED IN SERVICE:

RECORD OF MODIFICATIONS	

NOTES:

1. INFORMATION MAY BE PRINTED OR STAMPED THEREON, OR MAY BE AFFIXED BY LABEL IN ACCORDANCE WITH DDD-L-20, TYPE IV, CLASS 11.
2. NOMENCLATURE INCLUDES ITEM NAME AND TYPE DESIGNATION AS DEFINED IN MIL-STD-130.
3. SEE 3.3.3.7 FOR ADDITIONAL CONTAINER MARKING REQUIREMENTS.

FIGURE 3. Container/pack marking requirements for parachute identification.

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3.3.4 Electromagnetic radiation (EMR).

3.3.4.1 EMR hazard. For designs incorporating electrically-initiated explosive, propellant, or pyrotechnic subsystems or components, the parachute shall be designed to preclude hazards from electromagnetic radiation to ordnance (HERO) in accordance with MIL-STD-1385 (see 4.5.20.1).

3.3.4.2 Electromagnetic interference and electromagnetic compatibility (EMI/EMC). For designs incorporating electronic, electrical, or electromechanical subsystems or components, the parachute shall be designed to be compatible with the applicable aircraft EMI/EMC environment specified in MIL-E-6051, MIL-STD-461, and MIL-STD-462 (see 4.5.20.2).

3.3.4.3 Electromagnetic effects. For designs incorporating electronic, electrical, electro-pyrotechnic, or electro-mechanical systems or components, the parachute circuitry shall be designed to withstand, without damage or disruption, either one of the following (see 4.5.20.3):

a. An open-circuit voltage of 1500 ±50 volts (V), with a rise time of 20 nanoseconds (ns) maximum as driven by a 50-ohm source, applied across each connector pin and its return path for 125 ±5 ns.

b. A short-circuit current of 10.0 ±0.5 amperes (A), with a rise time of 20 ns maximum as driven by a 50-ohm source, injected through each connector pin and its return path for 125 ±5 ns.

3.3.5 Interchangeability and replaceability. Similar parts, subassemblies, and assemblies of the parachute shall be interchangeable or replaceable (see 6.3.16) in accordance with MIL-I-8500 (see 4.5.21).

3.3.6 Human performance/human engineering. The human performance/human engineering requirements for the parachute shall conform to MIL-STD-1472 to the extent applicable to the parachute and its intended environment. For parachutes incorporating pyrotechnic or ballistic components, engineering practices shall include those applicable to working with hazardous materials. Failure to perform an operation in the specified manner shall not pose a threat to human life. Personnel-to-equipment interface shall be optimized to achieve maximum effectiveness of personnel during handling, rigging, packing, operation, and control, and to minimize demands upon personnel resources, skills, training, and costs (see 4.5.22).

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3.3.7 Safety. (See 4.5.23.)

3.3.7.1 Health and safety criteria. The parachute shall not contain any part or component that provides adverse explosive, mechanical, or biological, including toxicological and carcinogenic, effects on the handler or aircrewmember when handled or used in accordance with approved procedures.

3.3.7.2 System safety program. The parachute shall be designed to meet the system safety requirements of MIL-STD-882. The contractor shall conduct a system safety program in accordance with MIL-STD-882 to ensure that adequate consideration is given to safety during all life-cycle phases of the program and to establish a formal, disciplined program to achieve system safety objectives. When specified in the contract or purchase order, the system safety program shall be documented in a plan which shall be approved by the contracting activity prior to use (see 6.2.2).

3.3.7.2.1 System safety hazard analysis. When specified in the contract or purchase order, a hazard analysis shall be performed and documented by the contractor to systematically identify and evaluate hazards, both real and potential, for their elimination or control (see 6.2.2).

3.4 GFP. Applicable GFP shall be specified in the associated specification sheet (see 6.4). The contractor shall be responsible for the integration of GFP into the parachute and for interfacing with GFP external to the parachute.

3.4.1 GFP conformance to specification. In the event of a conflict or inconsistency between the performance or other critical parameter requirements of the contract GFP specification and the delivered GFP component, the contractor shall immediately notify the contracting activity.

3.4.2 Changes to GFP. Changes to GFP to meet requirements peculiar to a specific application may be proposed by the contractor but shall be approved by the contracting activity prior to implementation.

3.5 Documentation. The design and all elements of the design shall be documented by the contractor as specified below and in the contract or purchase order.

3.5.1 Design approval drawings. When specified in the contract or purchase order, design approval drawings shall be prepared by the contractor in accordance with DOD-D-1000 (see 6.2.2).

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3.6 Logistics.

3.6.1 Maintenance. Both the nonsealed (see 6.3.15) and sealed (see 6.3.16) parachutes shall be designed for maintenance at the O and I levels in accordance with and as defined in OPNAVINST 4790.2 series and MIL-STD-2080 (see 3.2.15).

3.6.2 Supply. The parachute shall be capable of being supplied to using activities through the normal supply chain, either by component part, subassembly, or complete assembly, as determined by the Government.

3.6.3 Facilities and facility equipment. The parachute shall be designed for maintenance (inspection, rigging, packing, repairing, and modifying) in a parachute loft which meets the requirements of NAVAIR 13-1-6.2.

3.6.4 Personnel and training. Personnel and training requirements for maintenance of the parachute shall be in accordance with NAVAIR 13-1-6.2.

3.7 Working environment. The working environment shall meet the minimum requirements of MIL-STD-1695.

3.8 Workmanship. Workmanship shall be of the highest quality to ensure optimum performance, reliability, and service life. All components and assemblies shall be free from defects, such as burrs, cracks, dents, sharp edges, snags, snares, tears, loose or hanging threads, chipped paint, contamination, corrosion, evidence of excessive wear, and foreign material (see 4.5.24).

3.8.1 Cleanup. Prior to and after final assembly, all subassembly parts and the completely assembled parachute shall be thoroughly cleaned of loose threads and other foreign material. Solvents which emit toxic fumes or which may degrade any plastics or textiles used in the parachute shall not be used to clean metal parts.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein (see 6.2.1). Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government (see 6.2.1). The Government reserves the right to perform or witness any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All parachutes must meet all the applicable requirements of Sections 3 and 5. The inspections set forth in this specification and the associated specification sheet shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification or in the specification sheet shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all the requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. Special inspections:
   (1) Design evaluation (see 4.3.1).
   (2) Textile integrity evaluation (see 4.3.2).
b. Quality conformance inspections:
   (1) Field inspections (see 4.4.1.2).
   (2) Laboratory inspections (see 4.4.1.3).

4.2 Measuring and test equipment.

4.2.1 Calibration and maintenance. All measuring and test equipment used by the contractor for performance of the inspections specified herein shall be calibrated and maintained in accordance with MIL-STD-45662. When specified in the contract or purchase order, the contractor shall prepare a Calibration System Description in accordance with MIL-STD-45662 (see 6.2.2). The Calibration System Description shall be approved by the contracting activity prior to use.

4.2.2 Standard test equipment. The contractor shall provide and maintain inspection and test equipment of sufficient quality and quantity to permit the performance of the required tests and measurements under the conditions and to the accuracies specified herein. Unless otherwise specified herein, the magnitude of any error introduced by the test equipment shall not exceed 10 percent of the allowable tolerance of the requirement being measured. Any error introduced by the test equipment shall not be applicable toward widening the allowable tolerance of the requirement being measured.

4.2.3 Special test equipment and circuits. When special test equipment and circuits, other than as specified herein or supplied by the Government, are devised or commercially available equipment is employed, all test equipment, circuits, and methods shall meet the requirements of 4.2.1 and 4.2.2 and shall be approved by the contracting activity prior to use.

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4.2.4 Gauges. The contractor shall provide whatever gauges are necessary to ensure conformance to the dimensional requirements of this specification and the applicable drawings. When specified in the contract or purchase order, the contracting activity will furnish drawings of pertinent Government final inspection gauges (see 6.2.1). This procedure does not relieve the contractor of his responsibility for the design and manufacture of the gauges required for satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Government final acceptance gauges.

4.3 Special inspections.

4.3.1 Design evaluation. When specified in the contract or purchase order and prior to releasing the design for production, a complete design evaluation and design qualification plan shall be prepared and tests performed on engineering models by the evaluation group identified in the contract or purchase order (see 6.2.1), with consultation and cognizance throughout all phases by the contracting activity. Detailed test plans, describing the sequences, test conditions, and number of prototype models required for the complete evaluation, shall be prepared upon definition of the system design approach. To determine the adequacy of the design to meet the requirements of Sections 3 and 5, the design evaluation shall include, but shall not be limited to, the inspections of 4.4.

4.3.2 Textile integrity evaluation. When specified in the contract or purchase order, the integrity of the textiles used in the design shall be evaluated by the evaluation group identified in the contract or purchase order (see 6.2.1). Detailed test plans, describing the tests, test sequences, test conditions, and number of prototypes or textile samples required for complete evaluation, shall be prepared by the evaluation group. As a minimum, and unless otherwise specified in the contract or purchase order, the design evaluation shall include, but shall not be limited to, the following tests from FED-STD-191 (see 6.2.1):

a. Cloth: Test Methods 5104, 5122, 5134, and 5450.

b. Cordage: Test Method 6016.

c. Webbing and tape: Test Method 4108.

d. Thread: Test Method 4100.

4.4 Quality conformance inspections. Unless otherwise specified in the associated specification sheet, the requirements of Sections 3 and 5 shall be verified by the field inspections of Table III and the laboratory inspections of Table IV under the applicable conditions and provisions of 4.4.1 (see 6.4).

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TABLE III. Field inspections.

Inspection	Requirement paragraph	Method paragraph
Interface[1/]	3.1.3	4.5.1
Vertical descent velocity	3.2.1 (Table I, Item 1)	4.5.2.1
Horizontal velocity, hands-off mode	3.2.1 (Table I, Item 2)	4.5.2.2
Oscillation	3.2.1 (Table I, Item 3)	4.5.2.3
Time from initiaion of deployment to point of last full open	3.2.1 (Table I, Item 4)	4.5.2.4
Acceleration	3.2.1 (Table I, Item 5)	4.5.2.5
Rate-of-turn	3.2.1 (Table I, Item 6)	4.5.2.6
Flight control features	3.2.2	4.5.3
Manual operation force[1/]	3.2.3	4.5.4
Container/pack manual opening system	3.2.3.1	4.5.4.1
Canopy/risers release system[1/]	3.2.4	4.5.5
Inadvertent canopy/risers release	3.2.4.1	4.5.5.1
Canopy overload	3.2.5	4.5.6
Canopy deflation features	3.2.6	4.5.7
Harness[2/]	3.2.7	4.5.8
Twisted-line opening capability	3.2.8	4.5.9
Rigging and packing	3.2.11	4.5.12
Reliability[1/]	3.2.14	4.5.15
Maintainability[1/]	3.2.15	4.5.16

[1/] Shall be verified by both field and laboratory inspections.

[2/] Shall be performed only when the harness is included in the design.

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TABLE IV. Laboratory inspections.[1/]

Inspection	Requirement paragraph	Method paragraph
Interface[2/]	3.1.3	4.5.1
Manual operation force[2/]	3.2.3.2	4.5.4.2
Container/pack manual opening system[2/]	3.2.3.1	4.5.4
Canopy/risers release system[2/]	3.2.4	4.5.5
Ballistics	3.2.9	4.5.10
Weight	3.2.10.1	4.5.11.1
Dimensions/volume	3.2.10.2	4.5.11.2
Color	3.2.10.3	4.5.11.3
Rain	3.2.12.1	4.5.13.1
Salt fog	3.2.12.2	4.5.13.2
Sand and dust	3.2.12.3	4.5.13.3
Pressure/altitude cycling	3.2.12.4	4.5.13.4
High temperature/humidity	3.2.12.5	4.5.13.5
High temperature cycling	3.2.12.6	4.5.13.6
Temperature/humidity shock	3.2.12.7	4.5.13.7
Solar radiation	3.2.12.8	4.5.13.8
Fungus[3/]	3.2.12.9	4.5.13.9
Acceleration, shock,, and vibration	3.2.12.10	4.5.13.10
Depressurization	3.2.12.11	4.5.13.11
Service life and shelf/total life	3.2.13	4.5.14
Reliability[2/]	3.2.14	4.5.15
Maintainability[2/]	3.2.15	4.5.16
Materials, processes, and parts	3.3.1 to 3.3.1.10	4.5.17
Fabrication procedures, textile components	3.3.2 to 3.3.2.2.1	4.5.18
Nameplates and product marking	3.3.3 to 3.3.3.7	4.5.19
EMR hazard	3.3.4.1	4.5.20.1

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TABLE IV. Laboratory inspections.[1/] (Contd.)

Inspection	Requirement paragraph	Method paragraph
EMI/EMC	3.3.4.2	4.5.20.2
Electromagnetic effects	3.3.4.3	4.5.20.3
Interchangeability and replaceability	3.3.5	4.5.21
Human performance/human engineering	3.3.6	4.5.22
Safety	3.3.7	4.5.23
Workmanship	3.8 and 3.8.1	4.5.24
Packaging	3.2.16, Section 5	4.5.25

[1/] As specified in the test plan, parachutes shall be forwarded for field inspections after laboratory inspections.

[2/] Shall be verified by both field and laboratory inspections.

[3/] Shall be performed only when design includes fungus-nutritive materials.

4.4.1 Inspection conditions and provisions.

4.4.1.1 Quantity of parachutes. The quantity of parachutes to be manufactured and inspected per design shall be as specified in the contract or purchase order (see 6.2.1).

4.4.1.2 Field inspections. Unless otherwise specified in the contract or purchase order, the field inspections of Table III will be performed by the Government (see 6.2.1).

4.4.1.2.1 Field test plan. The Government will prepare the test plan for the field inspections of Table III using MIL-T-18303 as a guide. As a minimum, the test plan will include, as applicable:

a. Complete description of tests to be performed (whirl tower, sled, aircraft dummy drop, and live jump) and the sequence of tests.

b. All deviations from Table III.

c. Configuration of test items.

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d. Rigging and packing requirements.

e. Applicable pretest and inflight checks.

f. Communication systems.

g. Test data acquisition and recording methods.

h. Safety procedures and provisions.

i. Post-test handling requirements.

j. Post-test failure investigation procedures.

k. For live jump tests:

(1) Qualification requirements for parachutists.

(2) Provision for pre- and post-test oral and physical examinations for parachutists.

(3) Provision for adding weight to the harness system to meet the suspended load requirements, if required.

(4) Parachute-mounted instrumentation requirements.

(5) Provisions for approval by the Committee for Protection of Human Subjects.

4.4.1.2.2 Field test report. Upon completion of field testing, the Government will prepare a final test report using MIL-STD-831 as a guide. As a minimum, the report will include all applicable test data such as test conditions (temperature, speed, direction of wind, relative humidity, barometric pressure, density altitude, velocity at launch, and time of day for each event), test provisions, test equipment, test equipment configuration, and test results.

4.4.1.2.3 Recording of test data. Data measured on board the parachute shall be recorded using either (1) telemetry which transmits the data to a remote receiving station where it is recorded, or (2) a digital solid-state recorder carried aboard the parachute. When telemetry is used, the data shall be transmitted in digital form using pulse-code modulation (pcm) in accordance with the requirements of IRIG Document 106 and IRIG Document 118. When an on-board solid-state recorder is used, the data shall be stored in nonvolatile solid-state memory. The minimum data sample rate shall be 1000 samples per second for each channel of data. The minimum data word size shall be 10 bits. Transducers shall have a minimum passband of 0 to 200 hertz (Hz) and shall be free of resonances in the 0 to 300 Hz range. Filtering of the analog signal shall be used to prevent aliasing of digital data. No filtering shall be used that affects the minimum passband.

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4.4.1.2.3.1 Riser force measurement. When required, the force in each parachute riser shall be measured as a function of time by an appropriate force transducer. These forces shall be algebraically summed to produce total riser force as a function of time. Peak riser forces shall be determined from the total riser force data.

4.4.1.2.4 Trajectory data. When required, trajectory data shall be obtained by a tracking system approved by the contracting activity. The tracking system shall measure the position of the parachute as a function of time with respect to a known reference point. The velocity of the parachute shall be obtained from these data by a method approved by the contracting activity. Trajectory data alone shall not be used to obtain the velocity of the parachute until 10 seconds minimum have elapsed after the parachute reaches full open, unless it can be demonstrated to the satisfaction of the contracting activity that the sample rate and position accuracy of the data are adequate to produce accurate results during the opening process.

4.4.1.2.5 Photographic coverage. Photographic coverage shall include sequence stills and telescopic tracking via video or motion picture cameras, or both, with a capability of 100 to 400 frames per second. Motion picture, sequence still, and video timing marks shall be coordinated with the telemetry and trajectory data timing.

4.4.1.2.6 International Civil Aviation Organization (ICAO) Standard Atmosphere conditions. Standard atmosphere conditions shall be the sea level values of the ICAO (see 6.3.4). Rate-of-descent data shall be converted as specified below to ICAO Standard Atmosphere sea level conditions prior to the calculations of 4.5.2.1:

$$V_C = V_V \sqrt{\frac{\rho}{\rho_O}}$$

where:

V_C = corrected rate-of-descent

V_V = measured rate-of-descent

ρ_O = ICAO sea level density

ρ = density at altitude where V_V is measured

4.4.1.2.7 Wind conditions. Unless otherwise specified in the individual inspection method, surface wind speed during field testing shall be not greater than 15 knots.

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4.4.1.2.8 Classification of test results. Test results shall be classified or ranked into one of the following categories, as applicable:

a. No test. Test results shall be classified "no test" when it is determined by analysis that the test failure was caused by personnel actions or that a failure of the support equipment prevented the valid measurement of the parameter under test.

b. System success. Test results shall be classified "system success" when the parachute meets, without failure, all the performance requirements of this specification, the associated specification sheet, and the contract or purchase order.

c. System failure. Except as defined in a. above, test results shall be classified "system failure" when the parachute fails to meet any of the performance requirements of this specification, the associated specification sheet, and the contract or purchase order. Each failure shall then be further categorized as either catastrophic, critical, marginal, or negligible in accordance with the definitions of 3.2.14.3; for example, "system failure--critical."

NOTE: Because multiple tests may be included in each live jump, sled run, or air or whirl-tower drop, test results could be classified into more than one category; for example, a "no test" of descent rate due to a trajectory data failure, along with a "system failure--marginal" or "system success" for acceleration or force levels.

4.4.1.3 Laboratory inspections. Unless otherwise specified in the contract or purchase order, the laboratory inspections of Table IV shall be performed by the contractor in accordance with the approved test plan (see 4.4.1.3.1 and 6.2.1). When feasible, sequential inspections may be combined provided it can be demonstrated by the contractor that the applicable requirements of Sections 3 and 5 have been individually met. Additional inspections shall not be added unless authorized by the contracting activity.

4.4.1.3.1 Laboratory test plan. When specified in the contract or purchase order, the contractor shall prepare a test plan for the laboratory inspections of Table IV (see 6.2.2). The test plan shall specify test sequencing and shall include test procedures prepared in accordance with MIL-T-18303. The test plan shall be approved by the contracting activity prior to use. The contractor shall not deviate from the approved test plan unless authorized by the contracting activity.

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4.4.1.3.2 Inspection record. When specified in the contract or purchase order, the contractor shall prepare an inspection record for each parachute manufactured under a specified contract (see 6.2.2). All pertinent test data and the results of all inspections shall be recorded on the inspection record.

4.4.1.3.3 Inspection environment. Unless otherwise specified in the individual inspection method, the laboratory inspections of Table IV shall be performed under the conditions specified in 3.7.

4.4.1.3.4 Manufacturing aids and test fixtures. Unless otherwise specified in the contract or purchase order, all manufacturing aids and test fixtures required for performance of the laboratory inspections of Table IV shall be designed and fabricated by the contractor, and shall be approved by the contracting activity prior to use (see 6.2.1).

4.4.1.4 Parachute reuse. Unless otherwise specified in the contract, purchase order, or herein, at the contractor's discretion, a parachute may be reused for subsequent field and laboratory tests (see 6.2.1). A parachute shall not be reused for any test when greater than 2 percent of the canopy area has been repaired. A parachute or any component thereof which has been subjected to age life projection, structural strength, environmental, or dummy drop tests shall not be reused for live jump tests.

4.5 Inspection methods. To determine conformance to the requirements of Sections 3 and 5, the following inspections shall be performed in accordance with the conditions and provisions of 4.4, and as specified in the individual inspection method or as modified in the associated specification sheet (see 6.4). The contractor shall not deviate from the following inspection methods, as modified in the associated specification sheet or the approved test plan, unless authorized by the contracting activity.

4.5.1 Interface. The interface requirements of 3.1.3 shall be verified by documentation comparison, Government final acceptance gauges, and field tests, as required.

4.5.2 Performance. The performance requirements of 3.2.1 shall be verified by field tests and as defined below.

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4.5.2.1 Vertical descent velocity The vertical descent velocity (rate-of-descent) shall be measured during steady-state descent with no canopy manipulation (Table I, Item 1). A minimum of five separate tests shall be performed. All test data shall be converted to ICAO standards as specified in 4.4.1.2.6. For each test, a minimum of 100 data points (N) shall be used for the following calculations:

a. The mean corrected vertical velocity (V_c) (see 4.4.1.2.6) shall be calculated for each test as follows:

$$\bar{V}_c = \frac{\Sigma V_c}{N}$$

where $\bar{V}_c$ = individual instantaneous vertical velocity point.

b. A weighted mean rate-of-descent shall be calculated for all tests as follows:

$$\bar{\bar{V}}_c = \frac{\sum_{m=1}^{M} N_m \bar{V}_{cm}}{\sum_{m=1}^{M} N_m}$$

where M = the number of tests.

c. A weighted standard deviation, S, of the rate-of-descent of all tests shall be calculated as follows:

$$S = \left[\frac{\sum_{m=1}^{M} N_m (\bar{V}_{cm} - \bar{\bar{V}}_c)^2}{\sum_{m=1}^{M} N_m - 1} \right]^{1/2}$$

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d. A statistical t-test shall be used to demonstrate that V_c does not exceed the limit of Table I, Item 1, with a single-sided probability of occurrence of 90 percent, calculated as follows:

$$t = \frac{\bar{\bar{V}}_c - L}{S/\sqrt{M}}$$

where L = limit from Table I, Item 1.

4.5.2.2 Horizontal velocity, hands-off mode. The horizontal velocity in the hands-off mode (Table I, Item 2) shall be measured by deploying the parachute from a stationary location of sufficient height to ensure a descent duration of 10 seconds minimum, such as from a drop tower. Each test shall be comprised of two drops minimum and, for each subsequent drop, the parachute shall be rotated about its central axis 180 degrees from the previous drop position. Not greater than 15 minutes shall elapse between drops in each test. Not less than 10 tests shall be conducted at the minimum suspended weight of 150 +5, -0 lb and not less than 10 tests shall be conducted at the maximum suspended weight of 300 +0, -5 lb. Surface winds shall not exceed 5 ft/s during the tests. If the parachute exhibits any drive (the inherent tendency of a parachute to glide horizontally), it shall be positioned such that it will drive into the wind on the first drop and with the wind on the second drop.

4.5.2.3 Oscillation. Oscillation angle (or the deviation of the central axis of the parachute from the vertical), as measured in degrees, shall be measured by a trajectory data system (see 4.5.1.2.4) or other system approved by the contracting activity. Oscillation shall be measured during steady-state descent, hands-off mode, from 1000 feet above ground level (AGL) until ground contact. The specified oscillation limit of Table I, Item 3, shall not be exceeded more than twice during the 1000-foot measurement distance. The density altitude during testing shall be not greater than 5000 feet.

4.5.2.4 Time from initiation of deployment to the point of last full open. The time from initiation of deployment (instant of container/pack opening activation (see 6.3.6)) to the point of last full open (see 6.3.7) (Table I, Item 4) shall be measured at a true airspeed at initiation of 100 ±10 ft/s and a density altitude of 8000 ±500 feet.

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4.5.2.5 Acceleration. Acceleration during deployment (opening shock) shall be measured at a density altitude of 8000 ±500 feet and a velocity of 300 ±10 KEAS (Table I, Item 5). Acceleration shall be determined by dividing the total riser force data (see 4.4.1.2.3.1) by the suspended weight. For each test, acceleration shall be within the specified limits, except:

a. Accelerations with a duration less than 0.002 second may be disregarded.

b. The limits of Table I, Item 5, may be exceeded by 5 gravity units (g) for peak accelerations with a duration of 0.04 second or less.

4.5.2.6 Rate-of-turn. Rate-of-turn shall be measured in both the hands-off mode (see 3.2.2.1) and the controlled gliding mode (see 3.2.2.2 and Table I, Items 6a and 6b). The rate-of-turn shall be measured by timing three 360-degree turns in each mode and calculating the average.

4.5.3 Flight control features. The flight control features of 3.2.2 shall be verified during live jump tests. Horizontal velocity, in both the hands-off mode (see 3.2.2.1) and the controlled gliding mode (see 3.2.2.2), shall be measured using circular trajectory analyses. To perform tests for circular trajectory analysis, the parachute shall be trimmed to descend with a constant turn. The turn rate shall be from 5 to 12 degrees per second (deg/s). During the portion of a test for which circular trajectory data are obtained, the canopy shall not be manipulated in any way which could affect or control the turn rate, horizontal velocity, or rate-of-descent.

4.5.4 Manually-operated features.

4.5.4.1 Container/pack manual opening system. The container/pack manual opening system of 3.2.3.1 shall be verified during field tests by manually activating the system using one hand for one test and then the other for a subsequent test.

4.5.4.2 Manual operation force. The manual operation force of 3.2.3.2 shall be verified during laboratory tests by gradually applying a pull load until release occurs.

4.5.5 Canopy/risers release system. The canopy/risers release system of 3.2.4 shall be verified by manual operation of the system under a 300 +0, -5 lb suspended load. For the Type I parachute, the canopy/risers release system shall be in accordance with MIL-S-85076.

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4.5.5.1 Inadvertent canopy/risers release. The inadvertent release requirements of 3.2.4.1 shall be verified during field tests by releasing either the right or left side of a dual canopy/risers release system after full inflation of the main canopy.

4.5.6 Canopy overload. The canopy overload requirements of 3.2.5 shall be verified by field tests. With a 300 +0, -5 lb suspended load, the parachute shall be dropped at the airspeed required to achieve a total riser force of 6000 lbf minimum and a single riser force of 4500 lbf minimum.

4.5.7 Canopy deflation features. The canopy deflation features of 3.2.6 shall be verified by field tests over a water source.

4.5.8 Harness. When the harness is included in the design of the parachute, the harness requirements of 3.2.7 shall be verified by field tests. The test requirements shall be as specified in the associated specification sheet (see 6.4) and shall include a minimum of five drop tests per harness prior to the field tests of Table III. The following conditions shall apply:

a. The harness to be tested shall be constructed of all new textile and hardware components.

b. No repairs shall be made during the drop test series.

c. Any major change in design during testing (for example, stitch pattern variations, changes in textile or hardware components, rerouting of straps, relocation of hardware) shall invalidate all previous tests.

d. The risers, or harness attachment points, shall be attached to a snubbing cable which is attached to a fixed point on the drop tower. The forces in each riser shall be measured by a force transducer. The total load in the snubbing cable shall be measured by a single force transducer.

e. The harness shall be tested on a torso dummy weighing 300 +0, -5 lb. The dummy shall be dropped from a sufficient height to produce a force from 0 to 6500 lbf.

f. To measure webbing strap slippage through the hardware, each side of the webbing strap shall be marked with a contrasting color thread at a convenient reference point (for example, edge of a slide bar). After each drop test, the straps shall be examined for slippage.

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g. After the drop test series, the harness shall be examined for evidence of damage. Any hardware yield or damage to the primary load-carrying structure shall be considered a test failure.

4.5.9 Twisted-line opening capability. The twisted-line opening capability requirements of 3.2.8 shall be verified by field tests. The following conditions shall apply:

a. A minimum of five aircraft drop tests shall be conducted. Initiation of deployment shall be at 500 ±50 feet AGL, a velocity not greater than 100 KEAS, and with a 300 +0, -5 lb suspended load.

b. The suspension lines shall be twisted together 360 degrees three times within 30 inches beginning immediately below the deployment bag locking or emergence point. For the parachute assembly not incorporating a deployment bag, the twists shall begin immediately below the canopy skirt.

c. The parachute shall be rigged and packed in the approved manner.

NOTE: If difficulty is encountered in stowage of the twisted portion because of the increased girth, the line retaining member may be modified to obtain the desired retention characteristics, but shall be returned to its normal configuration prior to subsequent tests.

4.5.10 Ballistics. (See 6.5.)

4.5.10.1 Cartridge-actuated devices. The requirements of 3.2.9.1 for cartridge-actuated devices shall be verified in accordance with MIL-D-23615. The internal cartridges shall be verified in accordance with MIL-D-21625.

4.5.10.2 Electric initiators. The requirements of 3.2.9.2 for electric initiators shall be verified in accordance with MIL-I-23659.

4.5.10.3 Gas-actuated devices. The requirements of 3.2.9.3 for gas-actuated devices shall be verified in accordance with MIL-D-23615.

4.5.11 Physical characteristics.

4.5.11.1 Weight. The weight requirements of 3.2.10.1 shall be verified by weighing the parachute.

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4.5.11.2 Dimensions/volume. The dimension requirements of 3.2.10.2 shall be verified by measuring the physical dimensions of the parachute. The following method may be used to determine the volume of canopies up to 800 cubic inches (in^3) in size:

a. Except for deployment devices sewn to the canopy (for example, diapers) and the connector links and sliders on ram-air-inflated parachutes, strip the canopy of all extraneous items such as risers/harness, steering toggles, and deployment bags.

b. Fluff the canopy.

c. Using the parachute volume measurement fixture of Drawing SK 13-018-001-85, place the canopy into the fixture chamber, distributing the material as evenly as possible and as follows (see Figure 4):

(1) Place the round canopy into the chamber apex first, suspension lines last.

(2) Place the ram-air-inflated (or "square") canopy into the chamber tail first, suspension lines last.

d. Insert the piston into the cylinder, and manually press down on the piston to compact the canopy as much as possible.

e. Using weight bags or equivalent devices, apply enough weight to the top of the piston that, when combined with the weight of the piston, will place a total of 3 pound-force per square inch (psi) on the canopy, or a total of 210.00 ±0.25 lb calculated as follows: 70 square inches (in^2) of piston surface area x 3 psi = 210 lb.

f. Remove 140.00 ±0.25 lb from the top of the piston. Allow the remaining 70 lb to compress the canopy for an additional 30 seconds minimum.

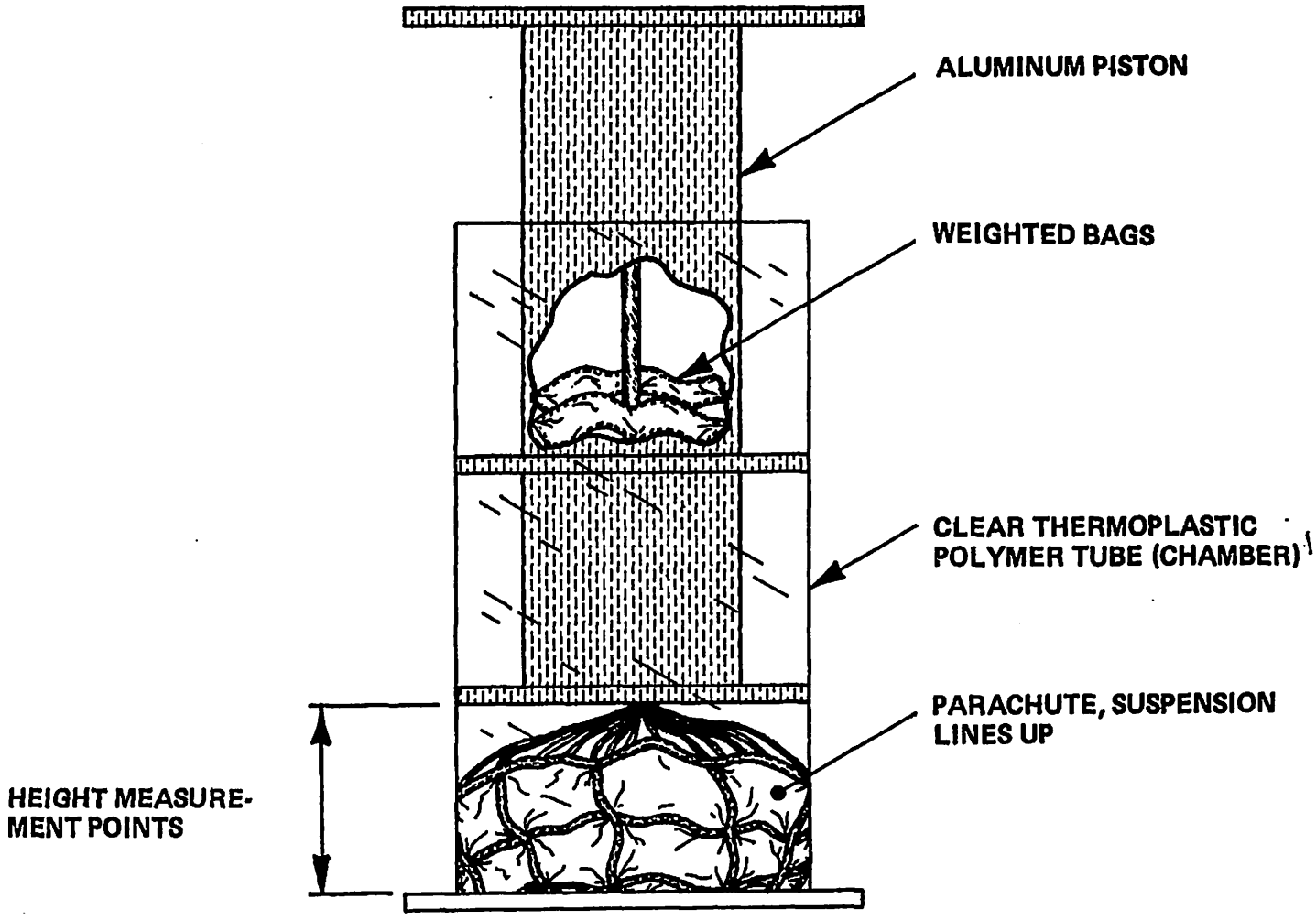
g. As shown on Figure 4, measure the height from the bottom of the cylinder to the bottom of the piston. Measurements shall be ±0.0625 inch.

h. Repeat steps b. through g. five times. Average the results.

4.5.11.3 Color. The color requirements of 3.2.10.3 shall be verified by visual examination.

4.5.12 Rigging and packing. The rigging and packing requirements of 3.2.11 shall be verified during field testing following certification.

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NOTES:

1. **Fixture shall be in accordance with Drawing SK 13-018-001-85.**
2. **See 4.5.11.2 for measurement procedure.**

FIGURE 4. Parachute volume measurement fixture.

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4.5.13 Environmental.

4.5.13.1 Rain. The requirements of 3.2.12.1 shall be verified in accordance with MIL-STD-810, Method 506.2, Procedure I, and as follows:

a.	Test item configuration	Packed condition.
b.	Rainfall rate	Equivalent to 0.8 mm/min.
c.	Droplet size	From 2.0 to 4.5 mm.
d.	Test item preheat temperature	Room ambient.
e.	Exposure	30 minutes per parachute container/pack side.
f.	Wind velocity	52 ±12 km/hr.
g.	Water temperature	Cycling from 5 to 45°C (40 to 115°F).

4.5.13.2 Salt fog. The requirements of 3.2.12.2 shall be verified in accordance with MIL-STD-810, Method 509.2, Procedure I, and as follows:

a.	Test item configuration	Packed condition.
b.	Test duration	48 hours.

4.5.13.3 Sand and dust. The requirements of 3.2.12.3 shall be verified in accordance with MIL-STD-810, Method 510.2, Procedures I and II, and as follows:

a.	Test item configuration	Packed condition.
b.	Test temperature	46 ±3°C (115 ±5°F).
c.	Relative humidity	30 +0, -5 percent.
d.	Air velocity, Procedure I	1750 ±100 ft/min.
e.	Air velocity, Procedure II	3540 to 5700 ft/min.
f.	Sand composition	Silica type.
g.	Sand concentration	0.0623 ±0.0150 g/ft^3.
h.	Dust composition	Red china type.
i.	Dust concentration	0.3 ±0.2 g/ft^3.

4.5.13.4 Pressure/altitude cycling. The requirements of 3.2.12.4 shall be verified by subjecting the parachute, in the packed condition, to pressure/altitude cycling beginning from 5 psia maximum for 1 minute minimum to 15 psia minimum for 1 minute minimum (one cycle) for 5000 cycles. Test apparatus shall be in accordance with MIL-STD-810, Method 500.2.

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4.5.13.5 High temperature/humidity. The requirements of 3.2.12.5 shall be verified by subjecting the parachute, in the packed condition, to a high temperature environment of +74 +0, -3°C (+165 +0, -5°F) at 5 percent RH maximum for 500 hours minimum and then 90 percent RH minimum for 500 hours minimum. Test apparatus shall be in accordance with MIL-STD-810, Method 501.2.

4.5.13.6 High temperature cycling. The requirements of 3.2.12.6 shall be verified by subjecting the parachute, in the packed condition, to a high temperature environment cycling from +20 to +74°C (+68 to +165°F) in 1 hour minimum, remaining at +74 +0, -3°C (+165 +0, -5°F) for 5 hours minimum, then back down to +20 +3, -0°C (+68 +5, -0°F) in 1 hour minimum and remaining at +20 +3, -0°C (+68 +5, -0°F) for 3 hours (one cycle) for 100 cycles. Test apparatus shall be in accordance with MIL-STD-810, Method 501.2.

4.5.13.7 Temperature/humidity shock. The requirements of 3.2.12.7 shall be verified by subjecting the parachute, in the packed condition, to temperature/humidity shock beginning at +74 +0, -3°C (+165 +0, -5°F) and 90 percent RH minimum for 2 hours minimum, then, within 1 minute, down to -54 +3, -0°C (-65 +5, -0°F) at 5 percent RH maximum for 2 hours minimum (one cycle) for 50 cycles. Test apparatus shall be in accordance with MIL-STD-810, Method 503.2.

4.5.13.8 Solar radiation. The requirements of 3.2.12.8 shall be verified by subjecting the parachute, in the packed condition, to the test of MIL-STD-810, Method 505.2, Procedure II, 60 cycles, except the radiant energy rate in Step 1 shall be 1120 W/m^2.

4.5.13.9 Fungus. The requirements of 3.2.12.9 shall be verified by subjecting the parachute, in the packed condition and then in the exposed condition, to the test of MIL-STD-810, Method 508.3, duration of 672 continuous hours.

NOTE: The test of 4.5.13.9 shall be performed only when the parachute contains fungus-nutritive materials.

4.5.13.10 Acceleration, shock, and vibration. The requirements of 3.2.12.10 shall be verified by subjecting the parachute, in the packed condition, to the acceleration, shock, and vibration levels specified in the associated specification sheet. Test apparatus shall be in accordance with MIL-STD-810, Methods 513.3, 514.3, and 516.3.

4.5.13.11 Depressurization. The requirements of 3.2.12.11 shall be verified by subjecting the parachute, in the packed condition, to rapid depressurization from 15 psia minimum to 1 psia maximum within 0.1 second.

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4.5.14 Service life and shelf/total life. The service life and shelf/total life requirements of 3.2.13 shall be verified in accordance with the age life projection test specified in the appendix.

4.5.15 Reliability. Laboratory and field tests shall be performed to verify the adequacy of the parachute to meet the minimum requirements of 3.2.14. These tests shall be preceded by a prediction in accordance with 3.2.14.2 to establish that the design-to-reliability requirement is theoretically possible. When specified in the contract or purchase order, test plans, test levels, sample size, and other details of the reliability demonstration shall be prepared by the contractor and approved by the contracting activity prior to use (see 6.2.2). Prior to the performance of any reliability tests, the reliability shall be predicted in accordance with MIL-STD-756 and verified to be equal to or in excess of the design goals of 3.2.14.

4.5.16 Maintainability. Unless otherwise specified in the contract or purchase order, the contractor shall provide a maintainability article for evaluation by the contracting activity to determine whether the parachute design meets all the maintainability requirements of 3.2.15 (see 6.2.1). All ballistic and pyrotechnic components of the prototype parachute maintainability article shall be inert; all mechanical firing mechanisms shall be operable. Applicable maintenance instructions and peculiar support equipment evaluation articles needed to accomplish and evaluate all maintenance requirements and actions at each maintenance level shall be provided with the maintainability article. The peculiar support equipment evaluation articles shall be of the proposed final design, but need not be manufactured by planned production processes. The maintainability article will be subjected to field maintainability review/evaluation by the contracting activity to:

a. Ensure system and component maintainability.

b. Develop information for evaluating contractor recommended maintenance procedures and practices.

c. Determine support equipment and peculiar support equipment suitability for performing the required maintenance tasks in a Fleet maintenance environment.

d. Type I. Evaluate the susceptibility of parachute components to damage during aircraft cockpit and recovery system (ejection seat) maintenance.

e. Type I. Determine adequacy of the design to meet the requirements of MIL-STD-2067.

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4.5.17 Materials, processes, and parts. The requirements of 3.3.1 through 3.3.1.10 shall be verified by visual examination, verification of conformance to applicable documents, and as specified in the contract or purchase order (see 6.2.1).

4.5.18 Fabrication procedures, textile components. The requirements of 3.3.2 through 3.3.2.2.1 shall be verified by visual examination, verification of conformance to applicable documents, and as specified in the contract or purchase order (see 6.2.1).

4.5.19 Nameplates and product marking. The marking requirements of 3.3.3 through 3.3.3.7 shall be verified by visual examination and verification of conformance to applicable documents.

4.5.20 EMR.

4.5.20.1 EMR hazard. The EMR hazard requirements of 3.3.4.1 shall be verified in accordance with MIL-STD-1385.

4.5.20.2 EMI/EMC. The EMI/EMC requirements of 3.3.4.2 shall be verified in accordance with MIL-E-6051, and the applicable procedures of MIL-STD-462 at the limits specified in MIL-STD-461.

4.5.20.3 Electromagnetic effects. The requirements of 3.3.4.3 shall be verified by subjecting the parachute to either one of the following electrical tests, as applicable:

a. An open-circuit voltage of 1500 ±50 V with a rise time of 20 ns maximum, as driven by a 50-ohm source, shall be applied across each connector pin and its return path for 125 ±5 ns.

b. A short-circuit current of 10.0 ±0.5 A with a rise time of 20 ns maximum, as driven by a 50-ohm source, shall be injected through each connector pin and its return path for 125 ±5 ns.

4.5.21 Interchangeability and replaceability. The interchangeability and replaceability requirements of 3.3.5 shall be verified in accordance with MIL-I-8500, and by visual examination and documentation reviews.

4.5.22 Human performance/human engineering. The human performance/human engineering requirements of 3.3.6 shall be verified in accordance with MIL-STD-1472.

4.5.23 Safety. The safety requirements of 3.3.7.1 and 3.3.7.2 shall be verified in accordance with MIL-STD-882.

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4.5.24 Workmanship. The workmanship requirements of 3.8 and 3.8.1 shall be verified by visual examination.

4.5.25 Packaging. The requirements of 3.2.16 and Section 5 shall be verified by visual examination and documentation comparison prior to shipment.

5. PACKAGING.

5.1 Preservation-packaging and packing. Unless otherwise specified in the contract or purchase order, preservation-packaging and packing shall be in accordance with FED-STD-102, Level C (see 6.2.1). Parachutes shall be packaged and packed in accordance with MIL-P-5610, Type I, and in a manner which will afford adequate protection during direct shipment from the supply source to the first receiving activity. Packing and containers shall conform to 49 CFR 171-178.

5.2 Marking. In addition to any special marking required by the contract or purchase order, all interior and exterior marking shall be in accordance with MIL-STD-129 (see 6.2.1).

6. NOTES.

6.1 Intended use. Emergency escape personnel parachutes are intended for use by aircrewmembers for emergency escape from an aircraft either via an ejection seat AAES (Type I) or by manual bailout (Type II) (see 1.1 and 1.2).

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Number, title, and date of this specification; number, title, and date of the applicable specification sheet; and whether parachute to be developed is Type I or Type II (see 1.2).

b. Design responsibility, if other than as specified in 3.1.2.

c. Group to perform validation/certification of rigging and packing procedures (see 3.2.11).

d. Requirements for special materials or parts, if applicable (see 3.3.1.1.a).

e. Requirements for identification and marking of parachutes to be used by agencies other than the U.S. Government, if applicable (see 3.3.3).

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f. Responsibility for inspections and inspection facilities, if other than as specified in 4.1.

g. Final inspection gauge drawings to be furnished by the Government, if applicable (see 4.2.4).

h. Provisions for design evaluation, and the independent evaluation group to prepare the design evaluation and design qualification plan, and to perform the design qualification tests (see 4.3.1).

i. Provisions for textile integrity evaluation, evaluation group to perform tests, and the applicable tests, if other than as specified in 4.3.2.

j. Quantity of parachutes to be manufactured per design (see 4.4.1.1).

k. Responsibility for performance of field inspections, if other than as specified in 4.4.1.2.

l. Responsibility for performance of laboratory inspections, if other than as specified in 4.4.1.3.

m. Requirements for design and fabrication of manufacturing aids and test fixtures, if other than as specified in 4.4.1.3.4.

n. Requirements for reuse of parachutes for subsequent tests, if other than as specified in 4.4.1.4.

o. Provisions for providing maintainability article for contracting activity evaluation (see 4.5.16).

p. Additional requirements for verification of materials, processes, and parts, if required (see 4.5.17).

q. Additional requirements for verification of textile components' fabrication procedures, if required (see 4.5.18).

r. Level of packaging and packing, if other than as specified in 5.1.

s. Special marking requirements for shipment, if required (see 5.2).

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6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered which incorporates a DD Form 1423, the data requirements identified below should be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL) incorporated into the contract. When the provisions of DOD Federal Acquisition Regulations (FAR) Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID	Options
3.2.11, 3.3.2, and 3.5.1	Engineering Drawings and Associated Lists	DI-E-7031	---
3.2.11	Rigging and Packing Procedures Certification/ Data Report	UDI-A-23264B	---
3.2.14.1	Reliability Program Plan	DI-R-7079	UDI-R-21131
3.2.14.2	Reliability Prediction and Documentation of Supporting Data	DI-R-7095	DI-R-7082
3.2.14.3	Failure Modes, Effects, and Criticality Analysis Report	DI-R-7085	---
3.2.15.3	Maintainability Program Plan	DI-R-7103	UDI-R-21131
3.3.7.2	System Safety Program Plan	DI-SAFT-80100	---
3.3.7.2.1	System Safety Hazard Analysis Report	DI-H-7048	---
4.2.1	Calibration System Description	DI-R-7064	---
4.4.1.3.1	Laboratory Test Plan	DI-T-5204	---
4.4.1.3.2	Inspection Record	UDI-M-20404	---
4.5.15	Reliability Test Plan	DI-R-7033	---

(DIDs related to this specification and identified in Section 6 should be approved and listed as such in DOD 5000.19L, Volume II, Acquisition Management Systems and Data Requirements Control List (AMSDL). Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.3 Definitions.

6.3.1 Contractor. The contractor is any corporation, company, association, or individual which undertakes performance under the terms of a contract, letter contract, letter of intent or purchase order, project order, and allotment, in which this document may be incorporated by reference. For the purpose of this specification, the term "contractor" also includes Government-operated activities undertaking performance by assigned airtask, project order, or allotment.

6.3.2 Prime system. The prime system is the system for which the parachute is an immediate subsystem; for example, the prime system for a Type I parachute is the ejection seat.

6.3.3 Contracting activity. The contracting activity is the Government procuring activity and the Government cognizant design activity.

6.3.4 Standard conditions. The standard conditions for this specification are:

a. Temperature 15°C (59°F)

b. Pressure 1.01325 bars

c. Density 1.2250 kilograms per cubic meter (kg/m^3)

6.3.5 Suspended load. The suspended load is all weight suspended below the risers.

6.3.6 Deployment. Deployment is that portion of a parachute's operation occurring from the instant of container/pack opening to the instant the suspension lines and main canopy are fully stretched, but prior to inflation of the canopy.

6.3.7 Point of last full open. The point of last full open is the first point where the projected area of the parachute exceeds the mean projected area of the fully open parachute minus one standard deviation of the projected area of the fully open parachute, and where no subsequent projected area falls below 75 percent of the mean projected area of the fully open parachute.

6.3.8 Density altitude. Density altitude is the altitude in the standard atmosphere corresponding to a particular value of air density.

6.3.9 Full control stroke. In the controlled gliding mode, the full control stroke is the minimum amount of movement of both sides of the control mechanism required to achieve a steady-state stall with a suspended load of 150 pounds. For parachutes where steady-state stall is prevented by the design, the full control stroke is the maximum amount of movement allowed by the design.

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6.3.10 Glide ratio. Glide ratio is the ratio of the horizontal velocity to the vertical velocity under zero wind conditions. Glide ratios for parachutes within the scope of this specification are categorized as follows:

a.	Low gliding parachute	Less than 1:1
b.	Medium gliding parachute	1:1 to 2:1
c.	High gliding parachute	Greater than 2:1

6.3.11 Packed condition. For the environmental requirements and inspections specified herein, the packed condition is the completely rigged parachute packed in its appropriate container, including, if applicable, protective coverings.

6.3.12 Exposed condition. For the environmental requirements and inspections specified herein, the exposed condition is the parachute out of its container and with as much component surface exposure as possible.

6.3.13 Service life. Service life of a parachute or component begins on the date when the parachute or component is first placed in service (or upon opening of manufacturer's package) and ends when a predetermined time period has elapsed.

6.3.14 Shelf/total life. Shelf/total life of a parachute or component begins on the date of manufacture and ends when a predetermined time period has elapsed.

6.3.15 Nonsealed parachute. A nonsealed parachute is a parachute not enclosed within an environmentally-sealed container.

6.3.16 Sealed parachute. A sealed parachute is a parachute enclosed in an environmentally-sealed container.

6.3.17 Interchangeability and replaceability. Interchangeability and replaceability are the substitution of similar parts, subassemblies, or assemblies without physical modification to any part of the component, including cabling, wiring, or mounting.

6.3.18 Abbreviations and symbols. Abbreviations and symbols used herein are defined as follows:

A	amperes
AAES	Aircrew Automated Escape System
AGL	above ground level

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AME	Aviation Structural Mechanic (U.S. Navy rating)
AMSDL	Acquisition Management Systems and Data Requirements Control List
°C	degrees Celsius
CDRL	Contract Data Requirements List
CFR	Code of Federal Regulations
deg	degrees
deg/s	degrees per second
DID	Data Item Description
DMMH/FH	direct maintenance man hour per flight hour
DOD	Department of Defense
DODISS	Department of Defense Index of Specifications and Standards
EMC	electromagnetic compatibility
EMI	electromagnetic interference
EMR	electromagnetic radiation
°F	degrees Fahrenheit
FAR	Federal Acquisition Regulations
FMECA	failure modes, effects, and criticality analysis
FSCM	Federal Supply Code for Manufacturers
ft	feet
ft/min	feet per minute
ft/s	feet per second
g	gravity units
GFP	Government-furnished property
g/ft^3	grams per cubic foot
HERO	hazards from electromagnetic radiation to ordnance

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hrs	hours
Hz	hertz
I	Intermediate level of maintenance
ICAO	International Civil Aviation Organization
in^2	square inches
in^3	cubic inches
IRIG	Inter-range Instrumentation Group
K	degrees Kelvin
KEAS	knots equivalent airspeed
kg/m^3	kilograms per cubic meter
km/hr	kilometers per hour
lb	pounds
lbf	pound-force
lbf/day	pound-force per day
m^3	cubic meters
max	maximum
min	minimum
mm	millimeters
$M_{max_{ct}}$	maximum corrective maintenance time
mm/min	millimeters per minute
MTTR	mean-time-to-repair
NAVAIR	Naval Air Systems Command
ns	nanoseconds
NWC	Naval Weapons Center, China Lake, CA 93555-6001
O	Organizational level of maintenance

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pcm	pulse-code modulation
PR	Aircrew Survival Equipmentman (U.S. Navy rating)
psi	pound-force per square inch
psia	pound-force per square inch absolute
RH	relative humidity
R/M	reliability/maintainability
sec	seconds
V	volts
W/m^2	watts per square meter
σ	standard deviation

6.4 Specification sheets. Associated specification sheets should specify the following, as applicable (see 3.1.1):

a. Number and title of this specification, and whether the parachute covered by the specification sheet is Type I or Type II (see 1.2).

b. All unique design/development requirements (see 3.1.1 and 3.2.1).

c. Interface control drawings and other engineering data required to define all physical and functional interfaces (see 3.1.3).

d. Operating temperature range, if other than as specified in 3.2.1.

e. Flight control features, if other than as specified in 3.2.2.

f. Horizontal velocity in the controlled gliding mode, if other than as specified in 3.2.2.2.

g. Manual operation (pull) force for applicable design features (see 3.2.3.2).

h. Maximum allowable weight (see 3.2.10.1).

i. Dimensions and volume requirements (see 3.2.10.2).

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j. Color scheme of parachute components, other than the canopy (see 3.2.10.3).

k. Color scheme of canopy, if other than as specified in 3.2.10.3.

l. Applicable acceleration, shock, and vibration levels for environmental testing (see 3.2.12.10 and 4.5.13.10).

m. Service life and shelf/total life requirements (see 3.2.13).

n. Materials and standard parts selection, if other than as specified in or in addition to 3.3.1.1.

o. Maximum allowable age of textiles (see 3.3.1.5.1).

p. Requirements for non-DOD electronic components, if applicable (see 3.3.1.6).

q. Manufacturing instructions, if other than as specified in 3.3.2.

r. Nameplates and product marking in addition to or if other than as specified in 3.3.3, 3.3.3.1, 3.3.3.3, and 3.3.3.7.

s. Marking ink, if other than as specified in 3.3.3.

t. Applicable GFP (see 3.4).

u. Verification of requirements, if other than as specified in 4.4.

v. Modifications and exceptions to the inspection methods of 4.5, if applicable.

w. Inspection requirements for the harness when included in the design (see 4.5.8).

6.5 Hazardous materials. Minimum safety precautions for use by the contractor are detailed in DOD Instruction 4145.26M, "DOD Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material."

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6.6 Subject term (key word) listing. The following key words apply for identification of the document during retrieval searches:

escape system, aircrew automated
parachute
parachute, bailout
parachute, ejection seat
parachute, emergency escape
parachute, personnel
recovery system, aircrew

Preparing activity:

Navy - AS
(Project 1670-N731)

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APPENDIX

AGE LIFE PROJECTION TEST

10.0 SCOPE.

10.1 Scope. This appendix establishes a test for determining the extent, rate, and rate equation of degradation, as a function of temperature, for the mechanical properties of materials. The test method allows for a more precise estimate of the age life and service life of new systems prior to Fleet deployment (see 4.5.14). This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

20.0 APPLICABLE DOCUMENTS.

This section is not applicable to this appendix.

30.0 PURPOSE.

30.1 Background. The aging of materials that results from the degradation of the materials' mechanical properties is caused by one or more chemical reactions. These chemical reactions occur primarily between the material and the surrounding air; active ingredients may include oxygen, oxides of sulfur and nitrogen, salt particles, and water vapor. These reactions are readily accelerated by increasing the temperature. The standard rule of thumb is that the reaction rate doubles for every 10°C (18°F) increase in temperature. In chemistry, the techniques for determining reaction rates and their rate equations have been known for decades and are published in numerous chemical kinetics text books. The basic relationship for the reaction rate is shown in equation (1).

$$\text{Rate} = k\,[\text{concentration}]^{r} \qquad (1)$$

where r is the reaction order and k is the rate constant designated by equation (2).

$$k = Ae^{-Ea/RT} \qquad (2)$$

where A is the frequency factor or probability of the meeting of the reacting species, Ea is the activation energy for the reaction, R is the

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gas constant, and T is temperature. The procedure for determining the rate equation is as follows:

a. Measure the concentration, [C], of the chemical reactants and products over a period of time, t, for the reaction at temperature, T.

b. Obtain the slope (rate of reaction) of the concentration versus time curve at each data point, graphically or mathematically.

c. Plot the logarithm of the rate of reaction (slope) versus the logarithm of the concentration, and calculate the line of best fit. The slope is equal to the reaction order, r.

d. Integrate the reciprocal of the concentration raised to the reaction order versus time, equation (3).

$$-\int_{[C]_o}^{[C]} \frac{1}{[C]^r}\, d[C] = \int_o^t k\,dt \qquad (3)$$

where [C] is the current concentration, $[C]_o$ is the initial concentration at t = 0, t is time, and k is the rate constant. The result is equation (4).

$$\frac{1}{[C]^{r-1}} - \frac{1}{[C]_o^{r-1}} = \frac{1}{r-1}\, kt \qquad (4)$$

e. Plot the relationship in equation (5).

$$\frac{1}{[C]^{r-1}} \text{ versus } t \qquad (5)$$

Calculate the slope, namely, the rate constant, k, of the resulting line.

f. Repeat a. through e. for several temperatures.

g. Plot the logarithm of the rate constant, k, versus temperature, T. The slope of the line is equation (6).

$$\text{slope} = -Ea/R \qquad (6)$$

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where Ea is the activation energy and R is the gas constant. The y-intercept is equal to the frequency factor, A (see equation (2)).

h. Calculate the activation energy, Ea, and determine the frequency factor, A.

i. Substitute all known values into equation (7).

$$\text{Rate} = Ae^{-Ea/RT}[C]^r \qquad (7)$$

30.2 Guidelines for determining test procedures and test conditions.

30.2.1 Application. This method shall be used when systems or components containing new or old materials of unknown aging resistance to the proposed use environment are being considered for Fleet use.

30.2.2 Restrictions. This method is not intended for age life projection of equipment, since too many variables are present to allow a reasonable chance of success (high risk). Material with changes in state (for example, glass transition temperature) within the chosen temperature test range may prove unacceptable for this method due to differences in reaction activation energies and rate constants on either side of the change-of-state temperature.

30.2.3 Sequence. Because this test requires destructive testing and new test samples, it cannot be performed with material subjected to previous testing. Only new, unused material shall be used. This test method must be performed separately from all other tests.

30.2.4 Choice of test temperature range. The test temperature range shall be chosen for its absence of change-of-state points for the material and an acceptable rate of degradation of the mechanical properties to be measured. If possible, the range shall include both the standard storage temperature and the use temperature range of the material, or at least be relatively close to them; this is to reduce the possible error resulting from the extrapolation of the data.

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APPENDIX

CAUTION

Do not choose a test temperature in close proximity to the melting point.

30.2.5 Choice of test temperatures. A minimum of five test temperatures between two adjacent change-in-state temperatures shall be chosen. For example, nylon 66 has a glass transition temperature at 50°C (122°F) and a melting point of 251°C (483°F). This is a 201°C (362°F) range. Dividing this range by the number of desired test temperatures gives a test temperature step of approximately 40°C (72°F). Thus, five acceptable test temperatures would be 50, 90, 130, 170, and 210°C (122, 194, 266, 338, and 410°F).

30.2.6 Duration of exposure. The material shall be exposed to the test temperatures for a sufficient length of time for the lowest test temperature to result in a minimum drop of 10 percent in the desired mechanical properties.

30.3 Special considerations.

30.3.1 Failure analysis. The failure of a material to meet the minimum age life requirements is dependent on information such as:

a. Maximum degradation allowed in the mechanical properties.

b. Degradation allowed in operating characteristics of the components and system.

If the material, component, or system specification does not designate a minimum allowable degradation in the critical mechanical properties of the material, a 20 percent maximum allowable degradation of the material's critical mechanical properties shall be used.

30.3.2 Summary of required test information. The following information is required in the test plan for adequate conduct of the tests of 40.0:

a. Test procedure.

b. Test duration to achieve a minimum of 20 percent degradation.

c. Temperature range.

d. Test temperatures.

e. Intended use of test item (environment/location).

f. Additional guidelines.

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APPENDIX

40.0 APPARATUS.

40.1 Test facility.

a. The required apparatus consists of a convection oven (or equivalent chamber or cabinet), and auxiliary instrumentation capable of maintaining and continuously monitoring the required temperature conditions throughout the envelope of air surrounding the test item.

b. The test item shall be totally surrounded by an envelope of air, except at the necessary support points. The temperature close to the test item shall be maintained within ±2°C (±3.6°F) of the test temperature and the temperature gradient within the chamber shall not exceed 1°C (1.8°F) per meter or a maximum of 2.2°C (4.0°F).

40.2 Controls.

a. Temperature. Unless otherwise specified in the test plan, if any action other than test item inspection or removal results in a change of the test item temperature or chamber air temperature of more than 5°C (9°F) for greater than 30 minutes in a 24-hour period, the test shall be stopped and restarted from the beginning using new samples, unless the effect is deemed insignificant by the project engineer.

b. Rate of temperature change. The temperature shall remain constant within ±2°C (±3.6°F).

c. Temperature measurement. Unless otherwise specified in the test plan, temperature sensor instrumentation is required for chamber air temperature only.

d. Temperature recording. Unless otherwise specified in the test plan, continuous recording of the chamber temperature is not required.

40.3 Test interruption.

a. Undertest interruption. An interruption which allows test temperatures to fluctuate outside allowable tolerances toward ambient conditions shall constitute a "no test." The test shall be repeated with new material samples.

b. Overtest interruption. Any interruption that results in more extreme exposure of the test item than required by the test plan shall constitute a "no test." The test shall be repeated with new material samples.

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c. Variations. The project engineer may waive the test interruption standards with the clear understanding that it will increase the data scatter and error. However, the test variation, the reason for it, and who authorized it must be clearly specified in the test report or inspection record.

40.4 Preparation for test.

40.4.1 Preliminary steps. Before initiating any testing, the following shall be determined from the test plan:

a. The test temperature range.

b. The test temperatures.

c. The test duration.

d. The extent of degradation required in the test item.

40.4.2 Pretest standard condition controls. Prior to the start of the test, determine the initial values of the mechanical properties of the material to be tested at standard conditions. For textiles, this is 21°C (70°F) and 65 percent RH. In most other cases, use 25°C (77°F) and 35 to 50 percent RH, unless the mechanical properties of the material are essentially unaffected by temperature and humidity (for example, metals).

40.5 Procedures. The following procedures provide the basis for collecting the necessary data for estimating the age life of the test item. All mechanical property testing shall be performed under the same conditions (temperature and humidity), unless the mechanical properties are unaffected by either temperature or humidity. In order to simplify the procedure, examples of each step using a fictitious material called "Samplelon" will be presented.

40.5.1 Data collection procedure.

Step 1. Determine the test temperature range, step, and values for the test item.

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Example: Samplelon has a glass transition temperature, Tg, at 40°C (104°F) and a melting point of 290°C (554°F) with no other changes-in-state between these two points. This gives a test temperature range of 40 to 290°C. If six test temperatures are desirable to cover the range, the test temperature step is

$$\text{step} = \frac{290-40}{6} = 41.67 \cong 40°C.$$

This gives the following test temperatures starting with the lowest range value: 40, 80, 120, 160, 200, 240°C (104, 176, 248, 320, 392, 464°F). If the rate of degradation at the lowest test temperature is excessively slow, this value may be eliminated as long as not less than five test temperature values remain. If not, choose a smaller range and another group of test temperatures.

Step 2. Adjust the temperature chamber to the desired test temperature and allow it to stabilize.

Step 3. Hang the test samples from their ends in the temperature chamber after it stabilizes at the test temperature. Do not allow the samples to touch the sides of the chamber or any other metal parts. Do not allow the chamber to drop more than 5°C (9°F). (A good technique is to hang the sample on a rack and place the rack quickly in the chamber.)

Step 4. Remove test samples at intervals designated in the test plan.

Example: For Samplelon, the test samples will be taken out of the chamber at 1, 3, 5, 7, 10, 15, 20, 25, 30, and 35 days of exposure.

Step 5. Allow the test samples to relax or stabilize at the standard test conditions (temperature and humidity) for 24 hours.

Step 6. Conduct a complete visual examination of the test samples and document the results.

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Step 7. Determine the critical mechanical properties of the test samples as designated in the test plan, and note any unusual occurrences during testing.

Step 8. Repeat steps 2 through 6 for each test temperature.

40.5.2 Age life projection procedure.

Step 1. Graph each of the mechanical properties versus time for test temperature, T, and draw the line of best fit.

Example: Samplelon was exposed to 160°C (302°F) for 45 days. Samples were taken at 1, 3, 5, 7, 10, 15, 20, 25, 30, and 35 days. The break strength of the material was 96.7, 90.7, 85.7, 81.2, 75.5, 68.0, 62.1, 57.3, 53.3, and 50.0 lbf, respectively. The break strength versus time is plotted on Figure 5.

Step 2. Obtain the slope (rate of property loss) of the mechanical properties versus time curve at each data point. The rate of property loss equals minus the slope.

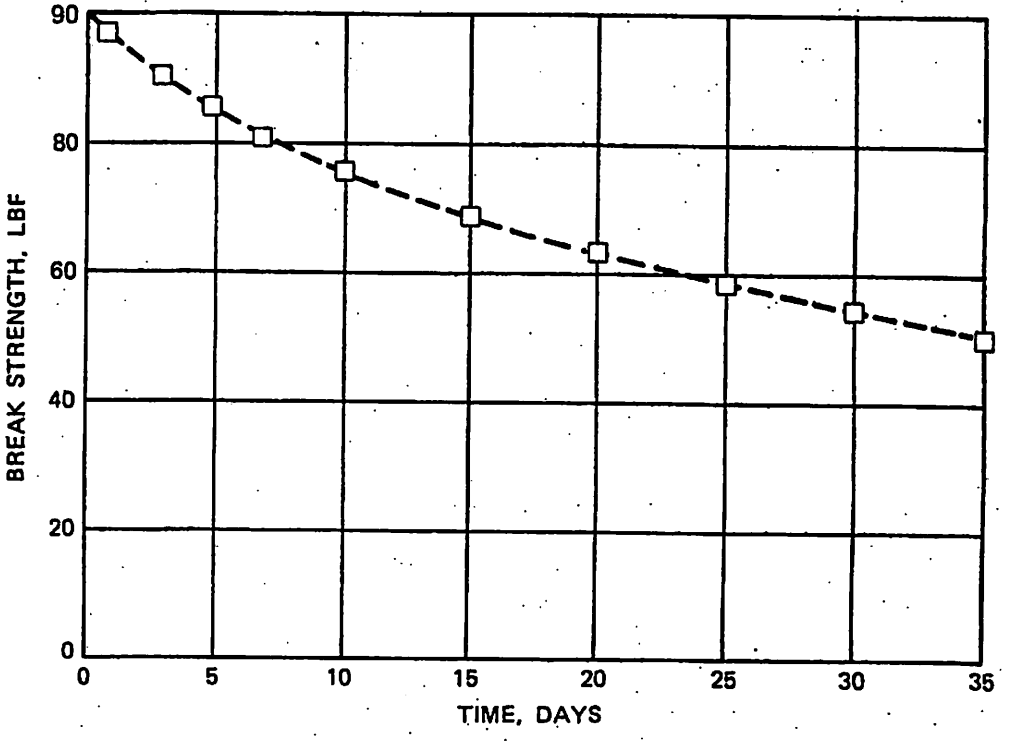
Example: Using Figure 5, measure from the graph the break strength values ±1 day of each data point and calculate the slope. For the day-1 point, the break strength values are 100 and 92.8 at 0 and 2 days. The slope is given by (see Table V for values):

$$\text{slope} = \frac{92.8 - 100}{2 - 0} = 3.6 \text{ lbf/day}$$

Step 3. Plot the logarithm of the rate of reaction (-slope) versus the logarithm of the mechanical property, and calculate the line of best fit. The slope is equal to the reaction order, r.

Example: Table V shows the $\log_{10}$ [rate] and $\log_{10}$ [B.S.] for Samplelon at 160°C (302°F). Figure 6 is the resulting plot. Because of the flattening out of the concentration versus time curve past 25 days, the 30- and 35-day slopes will not be used (error of measurement too high). Calculating line of best fit using linear regressions gives a slope (reaction order) of 2.5807. Round off to the nearest quarter, third, or half.

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1/ SEE 40.5.

FIGURE 5. Samplelon1/ break-strength loss at 160°C (302°F).

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TABLE V. Samplelon data at 160°C (302°F) from Figure 5.

Time, days	[B.S.] lbf	Axis	Initial	Final	Rate (slope), lbf/day	Log_{10} (rate)	Log_{10} [B.S.]
1	96.7	B.S. time	100.0 0	92.8 2	3.6	0.56	1.99
3	90.7	B.S. time	92.8 2	87.0 4	2.9	0.46	1.96
5	85.7	B.S. time	87.0 4	82.8 6	2.1	0.32	1.93
7	81.2	B.S. time	82.8 6	79.0 8	1.9	0.28	1.91
10	75.5	B.S. time	77.0 9	74.0 11	1.5	0.18	1.88
15	68.0	B.S. time	70.0 14	67.5 16	1.25	0.10	1.83
20	62.1	B.S. time	64.0 19	62.0 21	1.00	0	1.79
25	57.3	B.S. time	59.3 24	57.5 26	0.90	-0.046	1.76
30	53.3	B.S. time	55.0 29	53.3 31	0.85	-0.071	1.73
35	50.0	B.S. time	51.2 34	49.5 36	0.85	-0.071	1.70

Step 4. Calculate the reciprocal of $[P]^{r-1}$, where [P] is the mechanical property value and r is the reaction order determined in Step 3.

Example: For Samplelon, the term is

$$\frac{1}{[P]^{2.5-1}} = \frac{1}{[P]^{1.5}}$$

Table VI contains these values.

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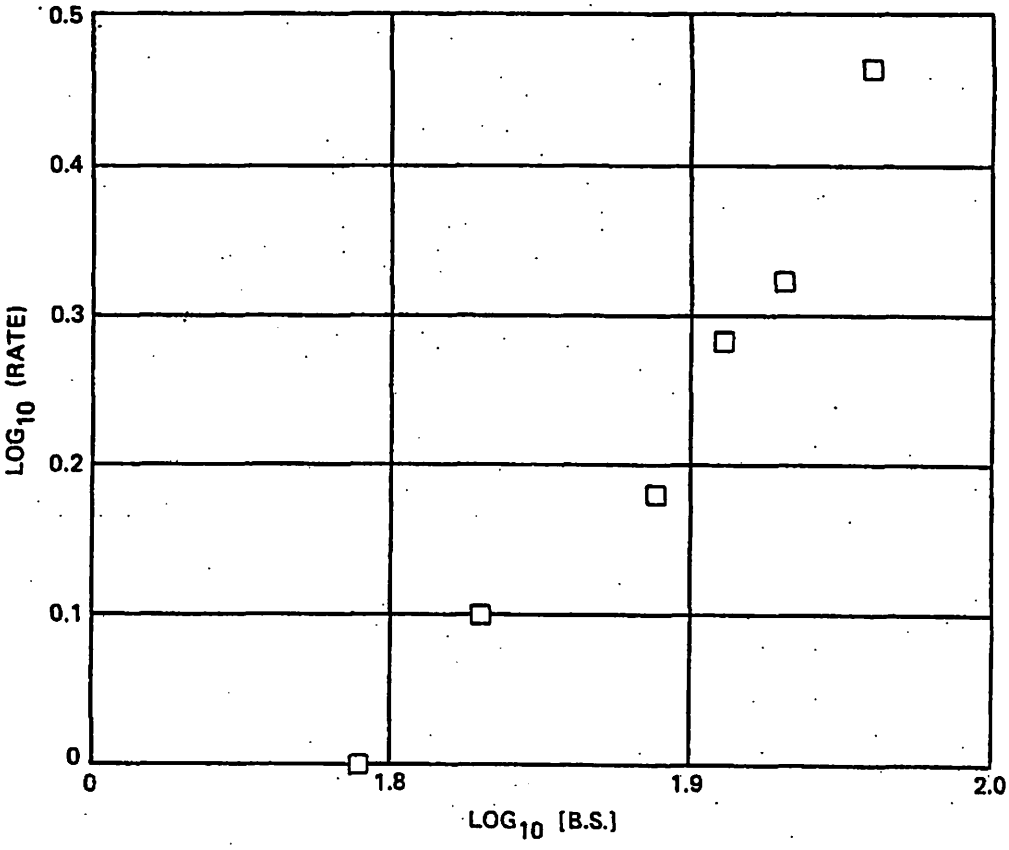


FIGURE 6. Plot of rate of property loss logarithim versus property value logarithim.

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Step 5. Calculate the slope of the following relationship using linear regressions (line of best fit) technique.

$$\frac{1}{[P]^{r-1}} \text{ versus time}$$

The slope is equal to the rate constant, k.

Example: Using the break strength, [B.S.], data from Table VI, the rate constant (slope) for Samplelon at 160°C (302°F) is 5.23×10^{-5}.

TABLE VI. Samplelon data for determining rate constant, k, at 160°C (302°F).

Time, days	[B.S.]	$1/[B.S.]^{1.5}$
1	96.7	0.00105
3	90.7	0.00116
5	85.7	0.00126
7	81.2	0.00137
10	75.5	0.00152
15	68.0	0.00178
20	62.1	0.00204
25	57.3	0.00231
30	53.3	0.00257
35	50.0	0.00283

Step 6. Repeat steps 1 through 5 for all test temperatures of the mechanical properties.

Step 7. Tabulate the rate constant data for each temperature.

Example: Table VII contains these data for Samplelon.

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TABLE VII. Samplelon rate constant data for mechanical property break strength.

Rate constant, k	ln k	Temperature, K	1/T
7.05×10^{-9}	-18.77	313	3.19×10^{-3}
2.70×10^{-7}	-15.12	353	2.83×10^{-3}
4.91×10^{-6}	-12.22	393	2.54×10^{-3}
5.23×10^{-5}	-9.86	433	2.31×10^{-3}
3.73×10^{-4}	-7.89	473	2.11×10^{-3}
1.96×10^{-3}	-6.23	513	1.95×10^{-3}

Step 8. Using the linear regressions (line of best fit) technique and the rate constant/temperature data, calculate the slope and the y-intercept of the natural logarithm of the rate constant versus reciprocal temperature.

Example: Using the data for Samplelon in Table VII, the slope = -10065 and the y-intercept = 13.39.

Step 9. Calculate the following relationships

$$\frac{-Ea}{R} = \text{slope}$$

$$A = e^{\text{y-intercept}}$$

Example: For Samplelon

$$\frac{-Ea}{R} = -10065$$

$$A = e^{13.39} = 6.53 \times 10^{5}$$

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Step 10. Substitute the calculated values into the following equation

$$\frac{1}{[P]^{r-1}} - \frac{1}{[P]_o^{r-1}} = (r - 1)Ae^{-Ea/RT}t$$

where $[P]_o$ is the initial undegraded property value, t is time, T is temperature in degrees Kelvin, A is the frequency factor calculated in Step 9, and -Ea/R is the negative activation energy divided by the gas constant calculated in Step 9. This gives you an equation of the mechanical property as a function of temperature and time. By using this equation, one can calculate the mechanical property value remaining after exposure to temperature, T, for time, t.

Example: For Samplelon

$$\frac{1}{[B.S.]^{1.5}} - \frac{1}{[100]^{1.5}} = 6.53 \times 10^5 \, e^{-10065/T}t$$

simplifying

$$[B.S.]^{1.5} = (6.53 \times 10^5 \, e^{-10065/T} \, t + 0.001)^{-1}$$

$$[B.S.] = (6.53 \times 10^5 \, e^{-10065/T} \, t + 0.001)^{-2/3}$$

If Samplelon is exposed to 160°C (302°F) for 20 days, what will its break strength be?

$$[B.S.] = (6.53 \times 10^5 \, e^{-10065/(160 + 273)} \, (20 \text{ days})$$

$$+ 0.001)^{-2/3} = 62 \text{ lbf}$$

From Table V, we see the experimental value is 62.1.

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NOTE: If the resulting equation of best fit from above is not a sufficiently accurate match to the experimental data, more terms representing additional reactions may be added to equation (7); for example:

$$\text{Rate} = A_e^{-Ea/RT}[C]^r + Be^{-Ea_2/RT}[C]^r + \ldots t$$

that, after integration yields

$$\frac{1}{[P]}r\text{-}1 - \frac{1}{[P]_o}r - 1 = (r - 1)\left(Ae^{-Ea_1/RT} + Be^{-Ea_2/RT} + \ldots\right)$$

In this case, Steps 4 through 10 of 40.5.2 will not work. However, Steps 1, 2, and 3 can be used to estimate the overall rate order, r. The rest of the constant will have to be determined by mathematical iteration methods using a computer system. These techniques are commercially available.

Step 11. Using the following equation, calculate the temperature increase

$$\Delta T_D = (3.413 \times 10^{-3} - 0.6931\ R/Ea)^{-1} - 293$$

in degrees Kelvin required to double the reaction rate, ΔT_D. Use the reciprocal of the Ea/R value calculated in Step 9.

Step 12. Using the standard cockpit worst day of Figure 7 for temperature and duration, the multiplication factor increase in the reaction rate relative to the reaction rate at 20°C (68°F), $M_{20°}$, is given by

$$M_{20°} = \left[\frac{1.5582\Delta T_D}{Tmax - Tmin}(D_1 + D_3)\frac{Tmax - Tmin}{2\ \Delta T_D} + 4\left(\frac{Tmax - Tmin}{2\ \Delta T_D}\right)^{-1}\right]$$

$$\left[\frac{\left(\frac{[P]_D}{[P]_o}\right) + 1}{2}\right]^r \Big/ 24\ \text{hrs}$$

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where ΔT_D is the temperature increase to double the reaction rate calculated in Step 11, Tmax is 74°C (165°F), Tmin is 20°C (68°F), D_1 is the duration of the temperature increase from 20 to 74°C (6 hrs), D_3 is the duration of the temperature decrease from 74 to 20°C (14 hrs), $[P]_D$ is the property value at maximum allowable degradation, $[P]_o$ is the initial property value, and r is the reaction order. For the standard cockpit worst day, the equation reduces to

$$M_{20} = \left[0.5771\ \Delta T_D(2^{54/\Delta T_D} - 1) + 4(2^{54/\Delta T_D})\right]\left[\left(\frac{[P]_D}{[P]_o} + 1\right)\Big/2\right]^r \Big/ 24$$

Example: For Samplelon, $\Delta T_D = 6.0335$, $r = 2.5$ $[P]_D = 80$, and $[P]_o = 100$.

then

$$M_{20} = \left[0.5771(6.0335)(2^{54/6.0335} - 1) + 4(2^{54/6.0335})\right]$$

$$\left[\frac{\frac{80}{100} + 1}{2}\right]^{2.5} \Big/ 24 = (1718.6 + 1978.3)\ \frac{0.7684}{24} = 118.4$$

The time to degrade at temperature, T, to 'x' of the initial property value is

$$t_T = \left(\frac{1}{(x[P]_o)^{r-1}} - \frac{1}{[P]_o^{r-1}}\right) \Big/ Ae^{-Ea/RT}$$

where t_T is the time to degrade at temperature, T, x is the fraction of property to remain after degrading to the allowable limit, $[P]_o$ is the initial property value, r is the reaction order, A is the frequency factor, T is temperature, Ea is the activation energy, and R is the gas constant. Except for x, these parameters are determined in Steps 1 through 10. An 80 percent retention of property value over the age life of a material is an acceptable default amount of degradation, if the amount of allowable degradation is not specified in the test plan or the material specifications.

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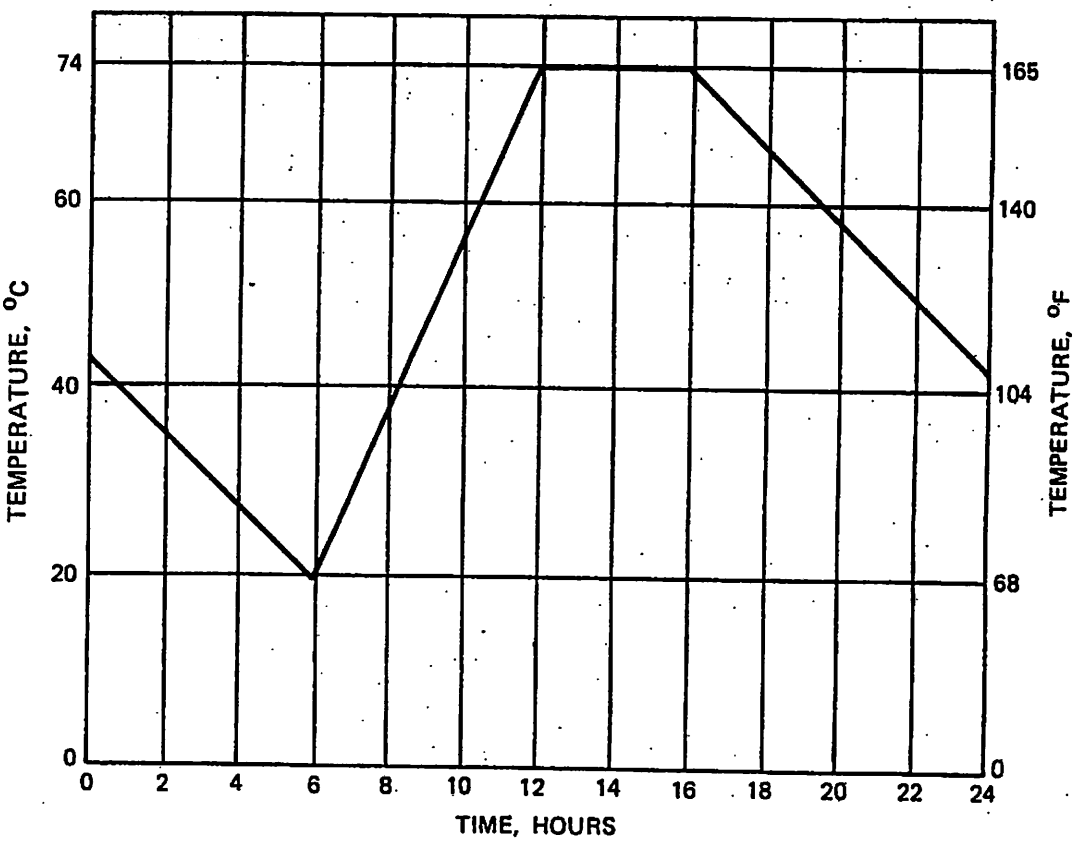


FIGURE 7. Standard cockpit worst day.

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Example: Substituting the values in for Samplelon at a reference temperature of 20°C (68°F), or 293 K, with x equal to 0.8

$$t_{20°} = \left(\frac{1}{(0.8[100])^{2.5-1}} - \frac{1}{[100]^{2.5-1}} \right) \Big/ 6.53 \times 10^{5}\, e^{-10065/293}$$

$$t_{20°} = 5.048 \times 10^{5} \text{ days}$$

Age life is given by

$$\text{Age life} = \frac{t_{20}}{M_{20}}$$

For Samplelon

$$\text{Age life} = \frac{5.048 \times 10^{5} \text{ days}}{118.4} = 4264 \text{ days or } 11.7 \text{ years.}$$

40.6 Information to be recorded.

a. Properties tested.

b. Property, time, and temperature data collected.

c. All information and graphs derived from the collected data.

d. All calculations required to obtain the estimated age life.

e. Estimated age life value.

f. Visual inspections and unusual occurrences.

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